When Does Jump-Risk Filtering Matter for Commodity Derivatives Deep Hedging? A Controlled Numerical Audit of Jump-Modulated No-Trade Bands, Price-Limit Locks, and Option-Gamma Spanning

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Abstract

Emerging strategic commodity futures—such as Lithium Carbonate (`LC`) futures on the Guangzhou Futures Exchange (GFEX)—exhibit self-exciting jump clustering alongside institutional daily price-limit bands (±7% to ±15%) that freeze order-book execution upon limit locks. This paper conducts a pre-registered, controlled semi-synthetic numerical audit (80 single-instrument Deep Hedging checkpoints, 60 Distributional Actor-Critic `D4PG-QR` runs, and 80 two-instrument Option-Gamma checkpoints across 100,000 S4 benchmark paths and 20,000 S5 extreme stress paths) to examine when and how causal online Hawkes jump-intensity filtering ($\hat{\lambda}_t$) and price-limit features improve tail-risk hedging under Conditional Value-at-Risk (CVaR_{95}). First, in the single-instrument futures benchmark ($c = 10^{-4}$), CVaR-trained Deep Hedging (`DH1`, $\mathrm{CVaR}_{95} = 0.021078 \pm 0.000105$) reduces tail loss by 27.00% relative to Black-76 delta hedging (0.028872) and by 32.64% relative to symmetric mean-variance hedging (`DH_mv`, 0.031294, $p_{\mathrm{holm}} = 3.9 \times 10^{-11}$), dominating mean-variance in the $(\mathrm{CVaR}_{95}, \mathrm{Cost})$ plane across $\lambda_{\mathrm{MV}} \in [0.5, 50.0]$ while avoiding the 20%–30% seed collapse rate of off-policy `D4PG-QR`. Second, embedding $\hat{\lambda}_t$ into a multiplicative no-trade band (`DH_lband`, `DH_full`) leads the network to learn a negative modulation slope ($\hat{\theta} = -0.0276 \pm 0.0202, p = 0.0019$ and $\hat{\theta} = -0.0230 \pm 0.0153, p = 0.0011$), narrowing the buffer during jump bursts and widening it during calm periods; this cuts trading frequency by 20.11%, saves 8.09% in transaction costs, reduces the cumulative lock-step hedging gap (`LockGap`) by 29.65% on S4 and 25.67% on S5 (where `DH_full` achieves the lowest $\mathrm{CVaR}_{95} = 0.020314$, −37.60% vs. `B76-Delta`), and matches unconstrained `DH1` under 6× transaction costs (0.023145 vs. 0.023241, $p = 0.218$) while saving 8.11% in friction. Third, input Jacobian saliency diagnostics (99.78%–100.00% weight on core states) and oracle-intensity experiments (`DH_oracle`, 0.021085) show that single-instrument futures hedging faces a spanning ceiling because linear futures ($\Gamma^{\mathrm{fut}} = 0$) cannot hedge quadratic jump convexity *ex ante*. Extending the action space to include a rolling 20-day ATM call option (`TorchHedgeEnvOpt`, 10 seeds) breaks this ceiling across all 2D policies (CVaR_{95} reduced by 17.98%–29.95% vs. 1D counterparts). While feeding $\hat{\lambda}_t$ as an unconstrained MLP input (`DH_opt1`, 0.015226 ± 0.001497) lowers 10-seed point CVaR_{95} by 11.91% vs. `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\mathrm{holm}} = 0.450$; +21.56% in the 5-seed subset) and overlaps with nonlinear σ_{EWMA} features (`DH_opt_ewma_nl`, 0.015904), channeling $\hat{\lambda}_t$ through the structural no-trade band (`DH_opt_lband`, $\hat{\theta} = -0.0290 \pm 0.0064, p = 1.59 \times 10^{-7}$) achieves the lowest realistic 2D tail risk of 0.014841 ± 0.001316, outperforming static-band `DH_opt_cband` (0.017546 ± 0.002298) by +15.41% ($t = -7.401, p_{\mathrm{holm}} = 0.000123$).

Keywords: Deep Hedging; Hawkes Jump Process; Online Intensity Filtering; Price-Limit Microstructure; Conditional Value-at-Risk (CVaR); Multi-Instrument Gamma Hedging; Lithium Carbonate Futures

1 Introduction and Related Work

1.1 Background and Research Motivation

The expansion of global green-energy supply chains has made strategic battery metals—such as Lithium Carbonate (`LC`) futures launched on the Guangzhou Futures Exchange (GFEX) in July 2023—important instruments for industrial risk management [26, 39] and cross-sector clean-energy portfolio hedging [13]. Unlike mature equity indices that approximate continuous geometric Brownian motion [4], emerging strategic commodity futures in China exhibit two major microstructural departures from frictionless diffusion [18, 34]. First, supply-chain bottlenecks, inventory squeezes, and policy announcements generate abrupt price discontinuities that cluster in time, a self-exciting process captured by Hawkes point models [5, 20] and marked jump-clustering specifications [11, 31]. Second, Chinese commodity futures exchanges enforce daily price-limit bands ($\pm 7\%$ to $\pm 15\%$ on GFEX `LC`) accompanied by directional order-book locks when prices reach the boundary [41, 42]. During a limit lock, one-sided liquidity vanishes, freezing a hedger's ability to rebalance a short option liability when underlying Gamma convexity losses are largest.

Classical analytical hedging rules—including Black [4] practitioner delta (`B76-Delta`), Leland [28] volatility-adjusted delta (`Leland-Delta`), and Whalley and Wilmott [40] asymptotic no-trade bands (`WW-Band`)—rely on local diffusive Taylor expansions and continuous trade execution. As a result, their hedging performance degrades under jump clustering and execution freezes [25, 38]. Introduced by Buehler et al. [7], **Deep Hedging (DH)** formulates derivative hedging as an end-to-end neural network control problem under convex risk measures such as Conditional Value-at-Risk (CVaR, Rockafellar and Uryasev 37). Subsequent work has extended Deep Hedging to long-term derivatives [9], rough volatility [23], quadratic hedging benchmarks [15], parameter ambiguity [29], equal risk pricing [10, 30], empirical equity index options [17, 32], neural no-transaction band networks [24], signature volatility models [1], backward dynamic programming with transaction costs [6, 19], and model-free distributional reinforcement learning (`D4PG-QR`, Cao et al. 8, Pickard and Lawryshyn 35).

Despite this progress, a clear research gap remains at the intersection of self-exciting jump filtering and emerging-market institutional trading frictions. Specifically, three questions remain open:

1. **When and through which architectural channel does online jump-intensity filtering matter ($\mathbf{H_0}, \mathbf{H_1}$)?** Does feeding a causal Hawkes jump-intensity estimate $\hat{\lambda}_t$ into a deep hedging policy reduce tail risk when hedging solely with linear futures, or does it require a second, convex option instrument ($\Gamma > 0$) and a structural no-trade band parameterization to extract *ex-ante* jump-Gamma protection without optimization variance?

2. **How do jump intensity and price limits reshape microstructural trading rules ($\mathbf{H_1}, \mathbf{H_3}$)?** Can a neural policy learn without sign constraints to modulate its no-trade band width $w_t(x_t)$ in response to $\hat{\lambda}_t$ and pre-position its hedge ratio ahead of price-limit locks?

3. **How do risk objectives ($\mathbf{H_2}$) and optimization methods (`BPTT` vs. `D4PG-QR`) interact with these frictions?** Does symmetric mean-variance hedging or model-free

distributional reinforcement learning remain effective under self-exciting jumps and price-limit truncations?

1.2 Related Literature

Our study relates to three strands of the quantitative finance literature:

1. Deep Hedging and Convex Risk Optimization Under Market Frictions. Buehler et al. [7] established Deep Hedging via end-to-end Back-Propagation Through Time (BPTT) under proportional transaction costs and convex risk measures [37]. Carbonneau [9] and Dai et al. [15] evaluated neural networks for long-term and quadratic hedging, while Horvath et al. [23] and Lütkebohmert et al. [29] studied rough volatility and parameter ambiguity. Marzban et al. [30] and Carbonneau and Godin [10] formulated equal-risk pricing and hedging across single and multiple hedging instruments. Empirically and architecturally, Mikkilä and Kanniainen [32] and François et al. [17] showed that deep hedging outperforms practitioner delta hedging on equity index options, Na and Wan [33] and Goudenège et al. [19] developed recurrent and backward hedging schemes for American options with transaction costs, and Imaki et al. [24] and Babaei [3] analyzed no-transaction band networks and fixed-plus-proportional cost buffers. Abi Jaber and Gérard [1] and Brugière and Turinici [6] studied signature volatility models and data-augmented model-free deep hedging, while Shakeel et al. [38] examined neural SDE hedging under regime-switching jump diffusions. To the best of our knowledge, none of these studies incorporate exchange price-limit locks or identify the spanning and architectural boundary between linear futures hedging and multi-instrument Gamma hedging under Hawkes jumps.

2. Distributional Reinforcement Learning for Option Greek Hedging. Building on quantile regression distributional reinforcement learning (`QR-DQN`, Dabney et al. 14), Cao et al. [8] applied Distributional Deep Deterministic Policy Gradient (`D4PG-QR`) to hedge option Gamma and Vega under transaction costs, an approach reviewed by Pickard and Lawryshyn [35] and extended to shortfall-aware implied-volatility hedging by Chen et al. [12]. While `D4PG-QR` avoids differentiating through the market simulator, its reliance on off-policy multi-step quantile Bellman bootstrapping has not been tested against direct pathwise BPTT in finite-horizon commodity markets with self-exciting jump clusters and price-limit truncations.

3. Jump Clustering and Institutional Price Limits in Commodity Futures. Recent studies document pronounced tail risk, volatility asymmetry, and cross-market spillovers in China's lithium carbonate industry chain [26, 39], clean-energy metal portfolios [13], and Chinese commodity futures [18, 34]. To model contagious price shocks, Brignone et al. [5] and Mercuri et al. [31] developed option pricing models with jump clusters and compound CARMA-Hawkes processes, while Chen et al. [11] and de Truchis et al. [16] modeled marked Hawkes jump clustering and fractional volatility persistence. Qiao et al. [36] and He et al. [22] analyzed jump components and self-exciting jumps in volatility forecasting and variance swap valuation. In addition, He et al. [21] and Jouini et al. [25] studied jump volatility forecasting and crisis-regime dynamic hedging, while Zhang et al. [42] and Xia et al. [41] examined asymmetric risk connectedness and session volatility under Chinese futures price-limit rules. Combining causal online Hawkes filtering with institutional price-limit execution freezes for dynamic derivative hedging remains unexplored.

1.3 Summary of Contributions

Evaluating 80 formal 1D Deep Hedging (`DH`) checkpoints, 60 `D4PG-QR` runs, and 80 formal 2D Option-Gamma checkpoints across 10 random seeds (`9001-9010`) on 100,000 S4 benchmark paths and 20,000 S5 extreme stress paths in a controlled semi-synthetic environment partially calibrated to GFEX Lithium Carbonate (`LC`) futures, we make four calibrated contributions:

- **Contribution 1 (Asymmetric CVaR Dominance in ($\mathbf{CVaR}_{95}, \mathbf{Cost}$) over Baselines and Mean-Variance, $\mathbf{H_2}$):** Under the S4 futures-only benchmark (100,000 paths, $c = 10^{-4}$), relative to classical Black-76 delta hedging (`B76-Delta`) and symmetric mean-variance deep hedging (`DH_mv`, $\lambda_{\text{MV}} = 10$), our CVaR-trained policy (`DH1`) improves out-of-sample CVaR_{95} from 0.028872 (`B76-Delta`) and 0.031294 ± 0.000722 (`DH_mv`) to 0.021078 ± 0.000105 ($p_{\text{holm}} = 3.9 \times 10^{-11}$) while reducing transaction costs from 0.5015×10^{-3} (`DH_mv`) to 0.3166×10^{-3} and dominating `DH_mv` in the $(\text{CVaR}_{95}, \text{Cost})$ plane across $\lambda_{\text{MV}} \in [0.5, 50.0]$, at the cost of a higher loss standard deviation $\text{Std}(L)$ (0.013391 in `DH1` vs. 0.012539 in 10-seed `DH_mv` at $\lambda_{\text{MV}} = 10$ and 0.012798 at $\lambda_{\text{MV}} = 50$) because asymmetric CVaR_{95} leaves upside convex hedging gains ($L_T < 0$) unpenalized.

- **Contribution 2 (Jump-Modulated No-Trade Band and Price-Limit Pre-Positioning, $\mathbf{H_1}$ & $\mathbf{H_3}$):** Under the S4 benchmark (100,000 paths) and the S5 extreme stress regime (20,000 paths), relative to the static-band policy (`DH_cband`) and unconstrained policy (`DH1`), our Hawkes-modulated band (`DH_lband`) and full microstructure-aware policy (`DH_full`) improve S4 CVaR_{95} from 0.021512 ± 0.000133 (`DH_cband`) to 0.021187 ± 0.000183 ($p_{\text{holm}} = 0.00063$), reduce S4 trading frequency from 99.95% (`DH1`) to 79.85% and transaction costs from 0.3166×10^{-3} to 0.2910×10^{-3}, compress the cumulative lock-step hedging gap (`LockGap`) from 0.935×10^{-2} (`DH1`) to 0.658×10^{-2} (`DH_full`, -29.65%) on S4 and from 2.162×10^{-2} to 1.607×10^{-2} (-25.67%) on S5, and match unconstrained `DH1` under $6\times$ transaction costs (0.023145 vs. 0.023241, $p = 0.218$) while saving 8.11% in friction, at the cost of a 0.52% higher CVaR_{95} on low-cost $1\times$ S4 ($0.021187 > 0.021078$) where the 2.56×10^{-5} fee savings are smaller than the band-induced tracking error.

- **Contribution 3 (Option-Gamma Spanning and Structural Band Activation of Jump Filtering, $\mathbf{H_0}, \mathbf{H_1}, \mathbf{H_2}$):** Under the 10-seed two-instrument futures-and-rolling-ATM-option hedging environment (`TorchHedgeEnvOpt`, $\tau_h = 20$ days) on S4, relative to 1D futures policies (`DH1` 0.021078, `DH_lband` 0.021187) and 2D static-band hedging (`DH_opt_cband` 0.017546 ± 0.002298), adding the option leg reduces CVaR_{95} by 17.98%–29.95% across all 2D configurations, while embedding $\hat{\lambda}_t$ into the multiplicative no-trade band (`DH_opt_lband`, $\hat{\theta} = -0.0290 \pm 0.0064, p = 1.59 \times 10^{-7}$) achieves the lowest realistic 2D CVaR_{95} of 0.014841 ± 0.001316 (+15.41% vs. `DH_opt_cband`, $t = -7.401, p_{\text{holm}} = 0.000123$; −29.95% vs. 1D `DH_lband`), at the cost of requiring a liquid short-dated option market, whereas feeding $\hat{\lambda}_t$ into an unconstrained 2D MLP (`DH_opt1`, 0.015226 ± 0.001497) yields an 11.91% 10-seed point improvement over `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\text{holm}} = 0.450$; +21.56% in the 5-seed subset) that does not achieve standalone statistical significance due to cross-seed non-convex optimization variance and information overlap with nonlinear σ_{EWMA} features (`DH_opt_ewma_nl`, 0.015904) and attenuates from +11.91% to +5.31% ($\kappa_{\text{iv}} = 2$) and +0.73% ($\kappa_{\text{iv}} = 5$) when option implied volatility co-jumps with $\hat{\lambda}_t$.

- **Contribution 4 (Optimization Stability of Direct BPTT over Distributional RL):** Under the 10-seed formal benchmark on S4 across training budgets of 3,000 to 20,000 updates, relative to model-free Distributional Actor-Critic with Quantile Regression (`D4PG-QR (C1)`), end-to-end differentiable `DH1` (BPTT) improves all-seed CVaR_{95}

from 0.144200±0.242296 (and non-collapsed seed mean 0.038934±0.020739) to 0.021078± 0.000105 while reducing the seed collapse rate from 20.0%–30.0% to 0.0% (0/80 runs) and cross-seed coefficient of variation from 50.02% to 0.50% (0.09% at 20,000 updates), at the cost of requiring an explicitly known, piecewise-differentiable market transition and accounting simulator rather than purely model-free interaction.

2 Market Microstructure Model and Online Hawkes Jump Filtering

2.1 Underlying Futures Dynamics with Self-Exciting Hawkes Jumps

We consider a discrete-time commodity futures market indexed by trading days $t \in \{0, 1, \ldots, T\}$, partially calibrated to the daily settlement prices of Lithium Carbonate (`LC`) futures on the Guangzhou Futures Exchange (GFEX) [26, 39]. Let $F_t > 0$ denote the futures price at day t, normalized to $F_0 = 1.0$. Without price-limit truncations, the latent log-return $r^*_{t+1} = \ln(F^*_{t+1}/F_t)$ follows a discrete-time jump-diffusion process driven by a self-exciting Hawkes counting process [2, 20]:

$$r^*_{t+1} = \left(\mu_d - \frac{1}{2}\sigma^2 - \lambda_t \bar{k}\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_{t+1} + \sum_{j=1}^{N_{t+1}} J_{t+1,j}, \quad \varepsilon_{t+1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{1}$$

where $\Delta t = 1$ trading day, $\sigma = 0.016144$ (0.0161 day$^{-1/2}$, 25.63% annualized) is the 5%-trimmed diffusive daily volatility estimated from 534 non-roll GFEX `LC` return observations, and $\mu_d = 0$ under the martingale futures measure with jump compensator $\bar{k} = \mathbb{E}[e^J - 1]$ [4].

To capture shock clustering in energy-transition commodity markets [5, 31] and derivative variance dynamics [16, 22], the daily jump count $N_{t+1} \mid \lambda_t \sim \text{Poisson}(\lambda_t)$ obeys the discrete-time Hawkes recursion [11, 20]:

$$\lambda_t - \mu = e^{-\beta}(\lambda_{t-1} - \mu) + \alpha N_t, \quad \lambda_{-300} = \mu, \tag{2}$$

where $\mu > 0$ is the baseline intensity, $\beta > 0$ is the exponential decay rate, and $\alpha \geq 0$ is the self-excitation magnitude. In our simulator (`simulator.py` and `trainer.py:make_sim`), each trajectory is preceded by a burn_in = 300-step warm-up ($300\beta = 60$ relaxation lifetimes) so that λ_0 starts from its stationary ergodic distribution rather than a deterministic cold start at μ; the simulator generates $T_{\text{sim}} = 61$ steps to span the $T = 60$-day option hedging horizon plus a one-step terminal index buffer. The stationary branching ratio and unconditional mean intensity are

$$n = \frac{\alpha}{1 - e^{-\beta}} < 1, \quad \bar{\lambda} = \mathbb{E}[\lambda_t] = \frac{\mu}{1 - n}. \tag{3}$$

In our S4 benchmark, $\bar{\lambda} = 0.04$ day^{-1}, $n = 0.3$, and $\beta = 0.20$ day^{-1} ($\mu = 0.028$, $\alpha \approx 0.05438$). Following Kou [27], jump sizes $J_{t+1,j}$ follow an asymmetric double-exponential law with upward probability $p_{\text{up}} = 0.4$ and decay rates $\eta_{\text{up}} = 40.0$ and $\eta_{\text{dn}} = 25.0$. In the GFEX `LC` sample, an event study of large daily jumps ($|r_t| > 2\sigma$, excluding locked days) reveals statistically significant post-jump continuation over horizons $h = 1, 2, 3, 4$ days, with cumulative signed returns of +0.45% (Newey-West $t = 2.09$), +0.86% ($t = 3.47$), +0.98% ($t = 2.88$), and +1.81% ($t = 2.67$). Fitting an exponential decay $\Delta C_h \propto e^{-\beta(h-1)}$ to the initial post-jump incremental returns ($h = 1, 2, 3$: +0.45%, +0.41%, +0.12%) yields $\hat{\beta}_{\text{OLS}} = 0.728$ day^{-1}, while the $h = 1$–4 continuation half-life ($t_{1/2} = \ln(2)/\beta \in [1.0, 4.0]$ trading days) implies an empirical decay range $\beta \in [0.17, 0.73]$ day^{-1} that brackets both our benchmark choice $\beta = 0.20$ day^{-1} ($t_{1/2} = 3.47$ days) and our Stage-2 fast-decay regime $\beta = 0.50$ day^{-1}. Although a finite-sample Monte Carlo

audit ($\mu = 0.02, \alpha = 0.10, \beta = 0.20$) confirms that truncation-corrected MLE (`M1`) reduces limit-censoring bias in μ by 60.4% (+0.00046 vs. +0.00117 under naive MLE `M0`) and in α by 78.9% (−0.00066 vs. +0.00314), the 534-day historical sample contains only 10 tail events above 3σ, which is too small for low-variance joint point estimation of $(p_{\text{up}}, \eta_{\text{up}}, \eta_{\text{dn}}, n)$. We therefore frame our study as a **partially calibrated, semi-synthetic numerical audit** in which $(\sigma, \ell_t, p_{\text{lock}}, \beta)$ are anchored to GFEX `LC` data and $(p_{\text{up}}, \eta_{\text{up}}, \eta_{\text{dn}}, n)$ are pre-registered scenario parameters audited across $n \in \{0.0, 0.3, 0.6\} \times \beta \in \{0.2, 0.5\}$.

2.2 Institutional Price-Limit Truncation and Execution Freeze Mechanics

Chinese commodity futures exchanges enforce daily price-limit bands (`limit_ratio` $\ell_t > 0$) relative to the prior settlement price F_t [41, 42]. Across the 542 trading days of our GFEX `LC` calibration window, 18 statutory exchange adjustments—triggered by pre-holiday margin and price-limit widenings ahead of major Chinese public holidays (Spring Festival, National Day, and Labor Day) and delivery-month progression rules under GFEX risk-control guidelines—generated eight limit tiers: 87 days at $\ell = 7\%$ (16.05%), 116 at 8% (21.40%), 9 at 9% (1.66%), 279 at 10% (51.48%), 7 at 11% (1.29%), 3 at 12% (0.55%), 37 at 13% (6.83%), and 4 at 15% (0.74%). For each simulated path, $\ell_t \equiv \ell$ is drawn at $t = 0$ from this empirical distribution:

$$\mathbb{P}(\ell = r) = \begin{cases} 0.1605, & r = 0.07, \\ 0.2140, & r = 0.08, \\ 0.0166, & r = 0.09, \\ 0.5148, & r = 0.10, \\ 0.0129, & r = 0.11, \\ 0.0055, & r = 0.12, \\ 0.0683, & r = 0.13, \\ 0.0074, & r = 0.15. \end{cases} \tag{4}$$

When the latent price $F^*_{t+1} = F_t \exp(r^*_{t+1})$ breaches $F_t(1 \pm \ell)$, the settlement price is truncated to the boundary:

$$F_{t+1} = \text{clip}\big(F^*_{t+1},\, F_t(1-\ell),\, F_t(1+\ell)\big), \tag{5}$$

triggering $\text{limit_hit}_{t+1} = \mathbb{I}(|F^*_{t+1}/F_t - 1| \geq \ell)$. Conditional on $\text{limit_hit}_t = 1$, an order-book lock ($\text{locked_up}_t = 1$ or $\text{locked_dn}_t = 1$) occurs with probability $p_{\text{lock}} = 1/7 \approx 0.1429$, matching the historical ratio of 1 full-day lock out of 7 limit-hit days (6 limit-hit episodes totaling 38.63% cumulative absolute return; daily hit rate 1.29%, daily lock rate 0.185%). Because $p_{\text{lock}} = 1/7$ is calibrated from a small count of extreme historical events, we also audit policy sensitivity across $p_{\text{lock}} \in \{0.0, 1/14, 1/7, 1/4\}$ and partial lock-day liquidity fills $\pi_{\text{fill}} \in \{0.00, 0.10, 0.30\}$ in Section 4.2.

An exchange limit lock imposes a directional execution freeze on rebalancing. Given prior position $\delta^{\text{exec}}_{t-1}$ and target $\delta^{\text{target}}_t \in [-\delta_{\max}, \delta_{\max}]$ ($\delta_{\max} = 1.5$), an upper-limit lock ($\text{locked_up}_t = 1$) prevents long buying, while a lower-limit lock ($\text{locked_dn}_t = 1$) prevents short selling:

$$\delta^{\text{exec}}_t = \begin{cases} \min\big(\delta^{\text{target}}_t,\, \delta^{\text{exec}}_{t-1}\big), & \text{if } \text{locked_up}_t = 1, \\ \max\big(\delta^{\text{target}}_t,\, \delta^{\text{exec}}_{t-1}\big), & \text{if } \text{locked_dn}_t = 1, \\ \delta^{\text{target}}_t, & \text{otherwise.} \end{cases} \tag{6}$$

Following the exact accounting in `torch_env.py`, we measure the resulting execution friction via the cumulative pathwise unexecuted lock-step hedging gap:

$$\text{LockGap} = \mathbb{E}\left[\sum_{t=0}^{T-1} \big|\delta^{\text{target}}_t - \delta^{\text{exec}}_t\big|\, \mathbb{I}\big(\text{locked_up}_t \vee \text{locked_dn}_t\big)\right]. \tag{7}$$

2.3 Causal Online Hawkes Intensity Filter Under Price-Limit Censoring

In practice, the latent jump intensity λ_t and jump counts N_t are unobservable [21, 36]. Standard threshold filters detect jumps via $|r_t| > k\sigma$, but tight price limits ($\ell = 0.07$ or 0.08) censor large jumps at the boundary. To address limit censoring while preserving causal adaptedness ($\hat{\lambda}_t \in \mathcal{F}_t$), we construct a limit-augmented online Hawkes filter:

$$\hat{\lambda}_t = \mu + A_t, \quad A_0 = 0, \quad A_t = e^{-\beta} A_{t-1} + \alpha \hat{I}_t, \tag{8}$$

where the causal jump indicator $\hat{I}_t \in \{0, 1\}$ combines a 3σ threshold ($k = 3.0$) with the observable limit-hit indicator:

$$\hat{I}_t = \mathbb{I}\Big(|r_t| > k\sigma \ \vee \ \text{limit_hit}_t = 1\Big), \quad k = 3.0. \tag{9}$$

We audit the confusion matrix of $\hat{I}_t$ on two complementary benchmarks: (i) the unit-test diagnostic benchmark `TH2b` ($30,029$ true jump days, `reports/TH2b_summary.md`) and (ii) the frozen out-of-sample test set `S4_test.npz` ($100,000$ paths $\times 60$ steps $= 6.0 \times 10^6$ path-days containing $236,488$ true jump days, where 97.3% of limit-hit days correspond to true latent jumps). On `TH2b` at $k = 3.0$, the filter triggers $5,895$ total detections composed of TP $= 4,138$ true positives and FP $= 1,757$ false positives (with FN $= 25,891$ missed moderate jumps where $|J| < 3\sigma = 0.0484$), yielding precision TP/(TP + FP) $= 70.20\%$, true-positive recall TP/(TP + FN) $= 13.78\%$ (whereas the ratio (TP + FP)/(TP + FN) $= 5,895/30,029 = 19.63\%$ represents the raw detection-to-true-jump count ratio), linear correlation $r(\hat{\lambda}_t, \lambda_t) = 0.4388$, and mean filtered intensity $\mathbb{E}[\hat{\lambda}_t] = 0.02465$ versus $\mathbb{E}[\lambda_t] = 0.04496$ (relative bias -45.17%). On the frozen `S4_test.npz` benchmark at $k = 3.0$, the filter triggers $48,883$ detections (TP $= 44,052$, FP $= 4,831$, FN $= 192,436$), achieving precision 90.12%, recall 18.63%, linear correlation $r = 0.4455$, and $\mathbb{E}[\hat{\lambda}_t] = 0.03019$ versus $\mathbb{E}[\lambda_t] = 0.03995$ (relative bias -24.44%). Varying the threshold across $k \in \{2.0, 2.5, 3.0, 3.5\}$ on `S4_test.npz` reveals a standard precision–recall trade-off: lowering k to 2.0 or 2.5 raises recall to 33.40% (TP $= 78,995$, FP $= 70,626$, precision 52.80%, $r = 0.4555$, bias -13.17%) and 24.80% (TP $= 58,638$, FP $= 13,704$, precision 81.06%, $r = 0.4853$, bias -21.81%) at the expense of frequent diffusive false alarms, whereas raising k to 3.5 increases precision to 91.50% (TP $= 33,535$, FP $= 3,114$) but lowers recall to 14.18% ($r = 0.3947$, bias -25.81%). In our multiplicative no-trade band architecture $w_t(x_t) = \text{softplus}(g_\phi(x_{t,-\lambda})) \exp(\theta \hat{\lambda}_t)$ (Section 3), any constant scale underestimation $\hat{\lambda}_t \approx c_\lambda \lambda_t$ ($c_\lambda \in [0.55, 0.76]$) is automatically absorbed by the unconstrained learnable slope θ via $\tilde{\theta} = \theta / c_\lambda$ and the intercept of g_ϕ. Evaluating both $\hat{\lambda}_t$ (`DH1`) and true λ_t (`DH_oracle`) separates residual filter noise from instrument-spanning limits.

3 Microstructure-Aware Deep Hedging Architecture

3.1 Liability Valuation, Wealth Accounting, and Multi-Instrument Extension

We consider a financial intermediary hedging a short European at-the-money (ATM) call option on the futures contract with strike $K = F_0 = 1.0$ and maturity $T = 60$ trading steps ($t = 0, \ldots, 60$). Following `option_env.py:fair_p0` and `pricing76.py:implied_sigma_from_price`, at $t = 0$ the option premium $P_0 = \widehat{\mathbb{E}}[\max(F_T - K, 0)]$ is priced at the empirical martingale expectation of the terminal payoff across the simulated ensemble, and the constant Black-76 implied volatility $\sigma_{\text{imp}} = C_{\text{B76}}^{-1}(F_0, K, T, P_0)$ is inverted from P_0 [4]. On the S4 benchmark ($100,000$ paths), this yields $P_0 = 0.058595$ and $\sigma_{\text{imp}} = 0.018822$ day$^{-1/2}$ (29.88% annualized); on the S5 extreme stress set ($20,000$ paths), $P_0 = 0.064603$ and $\sigma_{\text{imp}} = 0.020756$ day$^{-1/2}$ (32.95% annualized). Why does $\sigma_{\text{imp}} = 0.018822$ exceed the diffusive volatility $\sigma = 0.016144$ by

+16.59% on S4? Under the jump-diffusion process in Equation (1), total daily return variance is $\mathrm{Var}(r_t) \approx \sigma^2 + \bar{\lambda}\mathbb{E}[J^2]$; because P_0 equals the jump-inclusive martingale option value, inverting the Black-76 formula at P_0 maps the expected jump-variance contribution $\bar{\lambda}\mathbb{E}[J^2]$ into σ_{imp}. Consequently, $\Delta_{\mathrm{B76},t}(\sigma_{\mathrm{imp}})$ and $\Gamma_{\mathrm{B76},t}(\sigma_{\mathrm{imp}})$ represent practitioner jump-inclusive implied Greeks rather than pure diffusive Greeks. For $t < T$, the liability is marked to market at $V_t = C_{\mathrm{B76}}(F_t, K, T-t, \sigma_{\mathrm{imp}})$, settling at exact payoff $V_T = \max(F_T - K, 0)$ [10, 30].

1. Single-Instrument Futures Hedging (`TorchHedgeEnv`). With one-way proportional transaction cost rate $c = 10^{-4}$, cumulative trading cost C_t^{fut} and futures gain G_t^{fut} evolve from $\delta_{-1}^{\mathrm{exec}} = 0$ as

$$C_t^{\mathrm{fut}} = \sum_{s=0}^{t-1} cF_s \left|\delta_s^{\mathrm{exec}} - \delta_{s-1}^{\mathrm{exec}}\right| + \mathbb{I}(t=T)\, cF_T \left|\delta_{T-1}^{\mathrm{exec}}\right|, \quad G_t^{\mathrm{fut}} = \sum_{s=0}^{t-1} \delta_s^{\mathrm{exec}}(F_{s+1} - F_s). \tag{10}$$

Net portfolio wealth is $W_t = P_0 - V_t + G_t^{\mathrm{fut}} - C_t^{\mathrm{fut}}$, and normalized terminal hedging loss is $L_T = -W_T/F_0$.

2. Multi-Instrument Option-and-Futures Gamma Hedging (`TorchHedgeEnvOpt`). To test whether jump filtering requires a convex instrument to span jump-Gamma risk [10, 38], `TorchHedgeEnvOpt` (`torch_env_opt.py`) adds a rolling short-dated ATM call option with time-to-maturity $\tau_h = 20.0$ days (`TAU_H` $= 20.0$) and position bound $\eta_t^{\mathrm{target}} \in [-\eta_{\max}, \eta_{\max}]$ ($\eta_{\max} = 2.0$). At day t, a unit ATM call ($K_t^{\mathrm{roll}} = F_t$) opens at $V_{\mathrm{open},t} = C_{\mathrm{B76}}(F_t, F_t, \tau_h, \sigma_{\mathrm{imp}})$ (with sample mean $\overline{V}_{\mathrm{open}} = 0.033311 \approx 3.33\%$ of F_0 on S4) and closes at day $t+1$ at $V_{\mathrm{close},t+1} = C_{\mathrm{B76}}(F_{t+1}, F_t, \tau_h - 1, \sigma_{\mathrm{imp}})$, yielding one-step option PnL $\Delta V_t^{\mathrm{opt}} = V_{\mathrm{close},t+1} - V_{\mathrm{open},t}$ and Black-76 Gamma $\Gamma_{\mathrm{B76},t} = \phi(d_{1,t})/(F_t\sigma_{\mathrm{imp}}\sqrt{\tau_h})$. When the futures contract is limit-locked ($\mathrm{locked_up}_t \vee \mathrm{locked_dn}_t = 1$), option rebalancing is frozen ($\eta_t^{\mathrm{exec}} = \eta_{t-1}^{\mathrm{exec}}$). In the baseline controlled experiment, the option leg incurs proportional cost rate $c_{\mathrm{opt}} = c = 10^{-4}$ applied to the option premium $V_{\mathrm{open},t}$ (so the per-contract dollar cost is $c_{\mathrm{opt}}V_{\mathrm{open},t} \approx 0.033cF_t$), with total two-instrument cumulative cost and option gain given by

$$C_T^{\mathrm{tot}} = C_T^{\mathrm{fut}} + \sum_{s=0}^{T-1} c_{\mathrm{opt}} V_{\mathrm{open},s} \left|\eta_s^{\mathrm{exec}} - \eta_{s-1}^{\mathrm{exec}}\right| + c_{\mathrm{opt}} V_{\mathrm{open},T} \left|\eta_{T-1}^{\mathrm{exec}}\right|, \quad G_T^{\mathrm{opt}} = \sum_{s=0}^{T-1} \eta_s^{\mathrm{exec}} \Delta V_s^{\mathrm{opt}}. \tag{11}$$

Because constant σ_{imp} and $c_{\mathrm{opt}} = c$ isolate pure Gamma spanning from implied-volatility risk premiums and wider option bid-ask spreads, Section 4.4 explicitly audits policy robustness under option cost multipliers $c_{\mathrm{opt}}/c \in \{1\times, 5\times, 10\times\}$ and jump-driven dynamic implied volatility $\sigma_{\mathrm{imp},t} = \sigma_{\mathrm{imp},0}(1 + \kappa_{\mathrm{iv}}\hat{\lambda}_t)$ for $\kappa_{\mathrm{iv}} \in \{0, 2, 5\}$.

3.2 Information Sets, Policy Network, and Multiplicative Intensity-Modulated No-Trade Band

To isolate each microstructural component across Hypotheses $\mathbf{H_0}, \mathbf{H_1}, \mathbf{H_2}, \mathbf{H_3}$, we define nested causal state vectors $x_t \in \mathbb{R}^d$ [19, 33]:

1. **Core State Features (6 dimensions):** normalized time-to-maturity $\tau_t = (T-t)/T$, log-moneyness $\ln(F_t/K)$, prior executed hedge $\delta_{t-1}^{\mathrm{exec}}$, normalized wealth W_t/F_0, Black-76 delta $\Delta_{\mathrm{B76},t}$, and EWMA volatility $\sigma_{\mathrm{EWMA},t}$ ($\sigma_0 = 0.0161$, $v_t = 0.94v_{t-1} + 0.06r_t^2$).
2. **Baseline Limit Ratio (1 dimension):** $\ln(1 + \ell_t)$, included in all baseline states (`DH_nolam`, $d = 7$).

3. **Jump-Intensity Signal (1 dimension, $\mathbf{H_0}$):** filtered intensity $\hat{\lambda}_t$ (`DH1`, $d = 8$) or true intensity λ_t (`DH_oracle`, $d = 8$).

4. **Explicit Price-Limit Features (3 dimensions, $\mathbf{H_3}$):** lock indicator $\mathbb{I}(\text{locked_up}_t \vee \text{locked_dn}_t)$, consecutive locked days $\text{consec_lock_days}_t$, and raw limit ratio ℓ_t (`DH_limit` and `DH_full`, $d = 11$).

5. **Two-Instrument Option Features (2 dimensions in `TorchHedgeEnvOpt`):** prior option position η_{t-1}^{exec} and rolling Gamma $\Gamma_{\text{B76},t}$.

For direct policies without a no-trade band (`DH_nolam`, `DH1`, `DH_oracle`, `DH_limit`, `DH_mv`), a $[128, 128]$ ReLU MLP $f_\psi : \mathbb{R}^d \to \mathbb{R}$ maps x_t to the target hedge position:

$$\delta_t^{\text{target}} = \delta_{\max} \tanh\big(f_\psi(x_t)\big), \quad \delta_{\max} = 1.5. \tag{12}$$

To test Hypothesis $\mathbf{H_1}$ [24, 40], we introduce a multiplicative no-trade band architecture (`BandPolicy`, extending neural band networks Babaei 3, Imaki et al. 24). A $[128, 128]$ ReLU center network $f_\psi(x_t)$ outputs $c_t = \delta_{\max} \tanh(f_\psi(x_t))$, while a $[32, 32]$ ReLU band network $g_\phi(x_{t,-\lambda})$ (taking all state features *excluding* $\hat{\lambda}_t$, with output weights initialized to 0.0 and bias to -3.0) sets the baseline half-width. In `DH_cband`, $w_t(x_t) = \text{softplus}(g_\phi(x_{t,-\lambda}))$. In `DH_lband` and `DH_full`, the half-width is modulated by $\hat{\lambda}_t$ via an unconstrained learnable scalar $\theta \in \mathbb{R}$ (initialized at $\theta_0 = 0.0$):

$$w_t(x_t) = \text{softplus}\big(g_\phi(x_{t,-\lambda})\big) \cdot \exp\big(\theta\, \hat{\lambda}_t\big). \tag{13}$$

The target position projects $\delta_{t-1}^{\text{exec}}$ onto $[c_t - w_t(x_t),\, c_t + w_t(x_t)]$ via the piecewise-differentiable soft-projection operator:

$$\delta_t^{\text{target}} = \text{clip}\Big(\delta_{t-1}^{\text{exec}} + \text{soft_proj}\big(\delta_{t-1}^{\text{exec}} - c_t,\, w_t(x_t)\big),\, -\delta_{\max},\, \delta_{\max}\Big), \tag{14}$$

where $\text{soft_proj}(u, w) = -\text{ReLU}(u - w) + \text{ReLU}(-u - w)$. When $\theta < 0$, higher jump intensity $\hat{\lambda}_t$ contracts $w_t(x_t)$ to enforce tighter rebalancing during jump clusters, whereas lower $\hat{\lambda}_t$ widens $w_t(x_t)$ to save transaction costs. Testing whether unconstrained training yields $\hat{\theta} < 0$ across random seeds provides a direct test of microstructural adaptation.

3.3 Risk Objectives and End-to-End BPTT vs. Distributional RL Optimization

Following Buehler et al. [7] and Rockafellar and Uryasev [37], our primary objective minimizes the Conditional Value-at-Risk (CVaR_α, $\alpha = 0.95$) of L_T [6, 9]:

$$\min_{\Theta,\, b \in \mathbb{R}} \mathcal{J}_{\text{CVaR}}(\Theta, b) = b + \frac{1}{(1-\alpha)B} \sum_{i=1}^{B} \max\big(L_T^{(i)}(\Theta) - b,\, 0\big), \tag{15}$$

where Θ denotes policy parameters, $b \in \mathbb{R}$ is the jointly learned VaR_α threshold, and $B = 1024$ is the batch size. To test Hypothesis $\mathbf{H_2}$, we compare $\mathcal{J}_{\text{CVaR}}$ against the symmetric mean-variance objective (`DH_mv`, Dai et al. 15):

$$\min_{\Theta} \mathcal{J}_{\text{MV}}(\Theta; \lambda_{\text{MV}}) = \widehat{\mathbb{E}}\big[L_T(\Theta)\big] + \frac{\lambda_{\text{MV}}}{2} \widehat{\text{Var}}\big(L_T(\Theta)\big), \tag{16}$$

with $\lambda_{\text{MV}} = 10.0$ in the main benchmark and $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 10.0, 25.0, 50.0\}$ in our frontier sweep.

Because the accounting and price-limit rules are piecewise differentiable almost everywhere, our DH policies optimize $\mathcal{J}_{\mathrm{CVaR}}$ via end-to-end Back-Propagation Through Time (BPTT) using Adam (actor_lr $= 3 \times 10^{-4}$, gradient clip 5.0). To compare BPTT against the model-free distributional reinforcement learning approach of Cao et al. [8] and Pickard and Lawryshyn [35] (see also Chen et al. 12), we implement a Distributional Deep Deterministic Policy Gradient with Quantile Regression (D4PG-QR, C1-C9) critic with $M_q = 51$ quantile atoms trained via quantile Huber loss ($\kappa_h = 1.0$, Dabney et al. 14) on one-step Rockafellar-Uryasev rewards $r_t = -\big((L_t - b)^+ - (L_{t-1} - b)^+\big)$ ($\tau_{\mathrm{polyak}} = 0.005$, buffer $100,000$, batch 256).

4 Empirical Results and Microstructural Diagnostics

4.1 Calibration Data, Benchmark Policies, and Statistical Protocol

The diffusive volatility, price-limit tier distribution, conditional lock probability, and post-jump continuation half-life are calibrated to historical daily settlement prices of the generic front-month Lithium Carbonate (LC) futures contract on the Guangzhou Futures Exchange (GFEX) [26, 39] over the window 2023-07-24 to 2025-10-20 (542 trading days, yielding 534 non-roll daily log-returns). In this calibration sample, raw daily log-returns exhibit a standard deviation of 0.02038, excess kurtosis of 2.078, skewness of 0.113, a 99th-percentile absolute return of 0.0624, a median-absolute-deviation volatility estimate of 0.01465, and a 5%-trimmed diffusive volatility of $\sigma = 0.016144$ (0.0161 day$^{-1/2}$).

To prevent data snooping and selection bias [17, 32], our experimental pipeline enforces a strict separation between hyperparameter tuning, model training, checkpoint selection, and out-of-sample testing:

1. **Training, Validation, and Frozen Test Sets:** Learning rates (actor_lr $= 3 \times 10^{-4}$, critic_lr $= 10^{-3}$ searched over $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$), batch size (1024 paths for DH, 256 transitions for D4PG-QR), and architecture widths ($[128, 128]$) are locked using three independent tuning seeds (8001-8003). Formal policies are trained in PyTorch 2.x (float64 precision) on a 64-bit Windows workstation for $8,000$ gradient updates (with cross-budget sensitivity evaluated at $3,000$, $10,000$, and $20,000$ updates) and evaluated every 500 updates on a fixed validation set (val_set.npz, $10,000$ paths). For every run, the checkpoint achieving the lowest validation CVaR_{95} is frozen as best.pt.

2. **Benchmark (S4) and Out-of-Distribution Stress (S5) Markets:** All frozen checkpoints are evaluated once on the primary benchmark test set **S4** (S4_test.npz, $N = 100,000$ paths, $T = 60$ hedging steps; $n = 0.3, \beta = 0.2, \bar{\lambda} = 0.04$, empirical limit distribution; $P_0 = 0.058595$, $\sigma_{\mathrm{imp}} = 0.018822$ day$^{-1/2}$; pathwise jump incidence 83.03% [83,028 paths], limit-hit path incidence 16.51% [16,506 paths], locked-path incidence 3.00% [2,997 paths]) and zero-shot transferred without retraining to the out-of-distribution (OOD) crisis stress set **S5** (S5_test.npz, $N = 20,000$ paths; branching ratio doubled to $n = 0.6$ at fixed $\bar{\lambda} = 0.04$, tight uniform price limit $\ell = 0.07$, jump amplitude scale $\times 1.5$; $P_0 = 0.064603$, $\sigma_{\mathrm{imp}} = 0.020756$ day$^{-1/2}$; pathwise jump incidence 65.80% [13,159 paths], limit-hit path incidence 35.77% [7,153 paths], locked-path incidence 10.08% [2,016 paths]) [18, 25]. Note that in S5, doubling n from 0.3 to 0.6 at fixed $\bar{\lambda} = 0.04$ reduces the baseline immigration rate from $\mu = 0.028$ to $\mu = 0.016$ day^{-1} ($\mathbb{P}(N_{1:60} = 0) \approx e^{-60\mu} \approx 38.3\%$), lowering the fraction of paths with at least one jump from 83.03% to 65.80% while concentrating jumps into violent self-exciting clusters that more than double the limit-hit path incidence (35.77% vs. 16.51%) and triple the locked-path incidence (10.08% vs. 3.00%). To audit tail performance on paths where price limits bind, we report Cond. CVaR95 (Hit-Sub) and Cond. CVaR95 (Lock-Sub), defined as the conditional 95% CVaR (mean

of the worst 5% terminal losses) evaluated *within* the subset of paths experiencing at least one limit hit ($\mathcal{S}_{\text{hit}}$: $N = 16,506$ on S4, top-5% tail of 825 paths; $N = 7,153$ on S5, top-5% tail of 358 paths) or at least one limit lock ($\mathcal{S}_{\text{lock}}$: $N = 2,997$ on S4, top-5% tail of 150 paths; $N = 2,016$ on S5, top-5% tail of 101 paths), respectively.

3. **Classical Analytical and Heuristic Baselines:** We benchmark neural policies against four non-neural strategies: (i) `No-Hedge` ($\delta_t \equiv 0$); (ii) `B76-Delta` practitioner delta $\delta_t^{\text{target}} = \Phi(d_{1,t})$ [4]; (iii) `Leland-Delta` transaction-cost-adjusted volatility delta using modified variance $\tilde{\sigma}^2 = \sigma_{\text{imp}}^2\big(1 + \sqrt{2/\pi} \cdot \frac{2c}{\sigma_{\text{imp}}\sqrt{\Delta t}}\big)$ [28]; and (iv) `WW-Band` asymptotic no-trade band around $\Delta_{\text{B76},t}$ with half-width $w_t^{\text{WW}} = \big(\frac{3cF_t\Gamma^2_{\text{B76},t}}{2\gamma_{\text{WW}}}\big)^{1/3}$ calibrated at risk aversion $\gamma_{\text{WW}} = 0.25$ and $\gamma_{\text{WW}} = 1.0$ [40].

4. **Multi-Seed Statistical Testing and Minimum Detectable Effect (MDE):** Each neural specification in both the 1D and 2D benchmarks is trained across $M = 10$ independent random seeds (`9001-9010`) using unbiased sample standard deviations (`ddof=1`). Across our $m = 4$ pre-registered hypothesis family $\mathcal{H} = \{\mathbf{H_0}$ (`DH1` vs. `DH_nolam`), $\mathbf{H_1}$ (`DH_lband` vs. `DH_cband`), $\mathbf{H_2}$ (`DH1` vs. `DH_mv`), $\mathbf{H_3}$ (`DH_limit` vs. `DH1`)$\}$, we report cross-seed Mean $\pm$ SD, paired two-sided Student's t-tests, Wilcoxon signed-rank tests, 10,000-resample paired bootstrap 95% confidence intervals (CI), and Holm-Bonferroni step-down adjusted p-values (p_{holm}). We distinguish between our pre-registered 10% practical economic relevance threshold and the statistical minimum detectable relative effect size at two-sided $\alpha = 0.05$ and 80% power, $\text{MDE}_{80\%} = (t_{0.975,M-1} + t_{0.80,M-1}) \cdot \text{SD}(\Delta)/(\sqrt{M}\,\overline{\text{CVaR}}_{\text{base}})$, where $t_{0.975,9} + t_{0.80,9} = 2.262 + 0.883 = 3.145$ for $M = 10$ (4.110 for $M = 5$). In the 1D S4 benchmark ($M = 10$), tight cross-seed pairing yields $\text{MDE}_{80\%} \in [0.45\%, 1.80\%]$; in the 2D option-hedging benchmark ($M = 10$), $\text{MDE}_{80\%}$ is 6.55% for $\mathbf{H_1}$ (`DH_opt_lband` vs. `DH_opt_cband`, 4.18% on converged seeds) and 5.46% for $\mathbf{H_2}$ (`DH_opt1` vs. `DH_opt_mv`), but 28.76% for unconstrained $\mathbf{H_0}$ (`DH_opt1` vs. `DH_opt_nolam`) due to higher optimization variance in unbanded 2D networks. Under our pre-committed decision rule, a hypothesis is declared both statistically and economically activated only if $p_{\text{holm}} < 0.05$ and its relative CVaR_{95} reduction exceeds both 10% and $\text{MDE}_{80\%}$; under our pre-registered title-framing contingency rule, because unconstrained 2D jump filtering ($\mathbf{H_0}$, `DH_opt1` vs. `DH_opt_nolam`) yields +11.91% ($p_{\text{holm}} = 0.450 > 0.05$) across 10 seeds whereas structural band modulation ($\mathbf{H_1}$, `DH_opt_lband` vs. `DH_opt_cband`) achieves +15.41% ($p_{\text{holm}} = 0.000123 < 0.05$), the manuscript title and thesis are framed around conditional mechanism inquiry (*"When Does Jump-Risk Filtering Matter..."*) rather than unconditional unconstrained jump-filtering superiority.

4.2 Baseline Performance on S4 and Microstructural Validation of $\mathbf{H_1}$ and $\mathbf{H_2}$

Table 1 summarizes the out-of-sample hedging performance of all classical baselines and neural Deep Hedging specifications across the 100,000 paths of the S4 benchmark market ($c = 10^{-4}$). Consistent across all 10 random seeds, every CVaR-trained Deep Hedging policy reduces tail risk relative to the analytical benchmarks: `DH1` achieves an out-of-sample CVaR_{95} of 0.021078 ± 0.000105, representing a 27.00% reduction relative to `B76-Delta` (0.028872), a 26.88% reduction relative to `Leland-Delta` (0.028826), and a 29.37% reduction relative to `WW-Band` ($\gamma_{\text{WW}} = 0.25$, 0.029841).

Three primary empirical conclusions emerge from Table 1 and Figure 1:

1. **Statistical Confirmation of $\mathbf{H_2}$ (Asymmetric CVaR vs. Symmetric Mean-**

Table 1: Out-of-sample hedging performance on the S4 benchmark (100,000 paths, 10 random seeds `9001-9010`, $c = 10^{-4}$). Bold entries highlight best or key microstructural metrics.

Policy / Configuration	CVaR_{95} (Mean ± SD)	VaR_{95}	CVaR_{99}	Std(L)	Cost ($\times 10^{-3}$)	Turnover	Trade Freq.	Lock Gap ($\times 10^{-2}$)	Cond. CVaR_{95} (Hit-Sub)	Cond. CVaR_{95} (Lock-Sub)
`No-Hedge`	0.272131	0.195324	0.386982	0.091207	0.0000	0.0000	0.00%	0.000	0.294908	0.280819
`B76-Delta`	0.028872	0.018049	0.047693	0.010983	0.3344	3.2907	97.42%	0.677	0.049075	0.069395
`Leland-Delta`	0.028826	0.018039	0.047602	0.010991	0.3340	3.2860	97.45%	0.676	0.048992	0.069287
`WW-Band` ($\gamma_{WW} = 0.25$)	0.029841	0.018857	0.048804	0.011688	0.2400	2.3456	62.72%	0.623	0.050307	0.070028
`WW-Band` ($\gamma_{WW} = 1.0$)	0.036137	0.023888	0.057094	0.015920	0.1690	1.6315	34.85%	0.450	0.058638	0.076012
`DH_nolam` (No $\hat{\lambda}_t$)	0.021073 ± 0.000166	0.013791	0.036855	0.013483	0.3137	3.0830	99.95%	0.931	0.037057	0.059718
`DH1` (Filtered $\hat{\lambda}_t$)	0.021078 ± 0.000105	0.013779	0.036973	0.013391	0.3166	3.1123	99.95%	0.935	0.037077	0.059416
`DH_oracle` (True λ_t)	0.021085 ± 0.000164	0.013734	0.036963	0.013313	0.3152	3.0981	99.95%	0.934	0.037069	0.059465
`DH_cband` (Const. Band)	0.021512 ± 0.000133	0.014144	0.037043	0.013261	0.2650	2.5995	58.05%	0.926	0.037152	0.058799
`DH_lband` ($\hat{\lambda}_t$-Band)	$\mathbf{0.021187 \pm 0.000183}$	0.014002	**0.036610**	0.013468	**0.2910**	**2.8575**	**79.85%**	0.912	**0.036848**	**0.058691**
`DH_limit` (Limit State)	$\mathbf{0.021021 \pm 0.000088}$	**0.013653**	0.037003	0.013298	0.3165	3.1097	99.95%	1.113	0.037213	0.059824
`DH_full` (Full Model)	0.021131 ± 0.000190	0.013840	0.036892	0.013326	**0.2986**	**2.9312**	**80.76%**	**0.658**	0.037075	0.059492
`DH_mv` (Mean-Variance)	0.031294 ± 0.000722	0.018533	0.054044	**0.012539**	0.5015	5.0256	99.95%	1.405	0.051650	0.075797

Notes: For all neural policies, CVaR_{95} is reported as cross-seed Mean ± SD across $M = 10$ independent training seeds (`9001-9010`, sample standard deviation `ddof=1`); deterministic analytical baselines have SD = 0. Cross-seed SD in this 1D futures table is of order 10^{-4} (0.000088–0.000722), whereas in the 2D option-hedging table (Table 3) cross-seed SD is of order 10^{-3} due to joint optimization over (δ_t, η_t). Cost ($C_T^{\text{fut}}/F_0 \times 10^{-3}$) and Turnover ($\sum_{t=0}^{T-1} |\delta_t^{\text{exec}} - \delta_{t-1}^{\text{exec}}| + |\delta_{T-1}^{\text{exec}}|$) include terminal liquidation at $t = T$. `Cond. CVaR95 (Hit-Sub)` and `Cond. CVaR95 (Lock-Sub)` denote the conditional 95% CVaR computed over the worst 5% terminal losses *within* the subset of paths experiencing at least one price-limit hit ($N = 16,506$, tail size 825) or at least one limit lock ($N = 2,997$, tail size 150), respectively.

Variance): Optimizing the Rockafellar-Uryasev CVaR_{95} objective (`DH1`) reduces out-of-sample CVaR_{95} by 32.64% relative to the mean-variance policy `DH_mv` (0.021078 vs. 0.031294; paired difference $\Delta = -0.010216$, 95% bootstrap CI $[-0.010644, -0.009760]$, $t = -43.091$, $p = 9.7 \times 10^{-12}$, $p_{\text{holm}} = 3.9 \times 10^{-11}$, Wilcoxon $p = 0.00195$). In addition, `DH_mv` incurs 58.40% higher transaction costs (0.5015×10^{-3} vs. 0.3166×10^{-3}) and a turnover of 5.0256 because quadratic variance penalizes upside hedging windfalls ($L_T < 0$) and downside tail losses ($L_T > 0$) symmetrically, achieving a lower loss standard deviation $\text{Std}(L) = 0.012539$ vs. 0.013391 at the expense of severe right-tail shortfall.

2. **Statistical and Microstructural Confirmation of $\mathbf{H_1}$ (Jump-Modulated No-Trade Band and Baseline Cost Trade-Off):** Incorporating the multiplicative jump-intensity modulation $\exp(\theta\hat{\lambda}_t)$ in `DH_lband` statistically reduces CVaR_{95} relative to the unmodulated band policy `DH_cband` (0.021187 vs. 0.021512; paired difference $\Delta = -0.000325$, 95% bootstrap CI $[-0.000433, -0.000224]$, $t = -5.973$, $p = 0.00021$, $p_{\text{holm}} = 0.00063$, Wilcoxon $p = 0.00195$, +1.51% relative improvement vs. $\text{MDE}_{80\%} = 0.79\%$). At the same time, `DH_lband` achieves the lowest extreme tail loss $\text{CVaR}_{99} = 0.036610$, the lowest conditional limit-hit subset $\text{CVaR}_{95} = 0.036848$, and the lowest conditional locked-subset $\text{CVaR}_{95} = 0.058691$ across all single-instrument models, while reducing trading frequency by 20.11% (from 99.95% to 79.85%) and saving 8.09% in transaction costs (0.2910×10^{-3} vs. 0.3166×10^{-3}) relative to unconstrained `DH1`. However, at the baseline 1× futures fee ($c = 10^{-4}$), `DH_lband` exhibits a 0.52% higher total CVaR_{95} than unconstrained `DH1` ($0.021187 > 0.021078$, a net loss of +0.000109), because the 8.09% fee saving (2.56×10^{-5} of F_0) is smaller than the tracking-error loss (1.35×10^{-4}) incurred by wider no-trade buffers during low-intensity periods; thus, in a 1D futures market, the jump-modulated band is a friction-saving mechanism whose net CVaR_{95} benefit materializes only when unit transaction costs rise to $3 \times$ –$6 \times c_{\text{real}}$ or when combined with a higher-spread option leg (Section 4.4). Inspecting the unconstrained learned parameter θ across the 10 seeds (`ddof=1`) reveals a cross-seed mean of $\hat{\theta} = -0.0276 \pm 0.0202$ (9/10 seeds negative, $t = -4.330, p = 0.0019$) for `DH_lband` and $\hat{\theta} = -0.0230 \pm 0.0153$ (10/10 seeds negative, $t = -4.743, p = 0.0011$) for `DH_full`. Without prior sign restrictions, gradient descent consistently learns $\theta < 0$, contracting the no-trade buffer during jump-clustering bursts to prioritize tail protection and widening it during tranquil periods to conserve transaction costs.

3. **Pre-Positioning Under Price-Limit Locks and Robustness Across p_{lock} and π_{fill}:** While `DH_limit` attains the lowest point CVaR_{95} (0.021021 ± 0.000088; paired difference vs. `DH1` $\Delta = -0.000057$, 95% CI $[-0.000105, -0.000010]$, $t = -2.357, p = 0.0428, p_{\text{holm}} = 0.0856$), combining explicit price-limit features with the intensity-modulated band in `DH_full` reduces the cumulative unexecuted lock-step hedging gap (`LockGap`) from 0.935×10^{-2} (`DH1`) to 0.658×10^{-2}—a 29.65% reduction on limit-locked days—while cutting transaction costs by 5.69% (0.2986×10^{-3} vs. 0.3166×10^{-3}). Auditing sensitivity across conditional lock probabilities $p_{\text{lock}} \in \{0.0, 1/14, 1/7, 1/4\}$ confirms that as p_{lock} increases from 1/14 (1.54% locked paths) to 1/7 (2.78%) and 1/4 (4.98%), `DH1`'s `LockGap` rises from 0.469×10^{-2} to 0.882×10^{-2} and 1.676×10^{-2}, whereas `DH_full` compresses `LockGap` to 0.331×10^{-2} (−29.56%), 0.625×10^{-2} (−29.09%), and 1.180×10^{-2} (−29.62%), respectively, while lowering locked-subset CVaR_{95} from 0.053910 to 0.052680 ($p_{\text{lock}} = 1/14$) and from 0.054302 to 0.053984 ($p_{\text{lock}} = 1/4$). Similarly, relaxing the complete execution freeze on locked days to allow partial order fills $\pi_{\text{fill}} \in \{0.00, 0.10, 0.30\}$ reduces `DH1`'s `LockGap` from 0.879×10^{-2} to 0.788×10^{-2} and 0.609×10^{-2}, while `DH_full` maintains an invariant 28.80% (0.626×10^{-2}), 28.74% (0.562×10^{-2}), and 28.59% (0.435×10^{-2}) relative `LockGap` reduction.

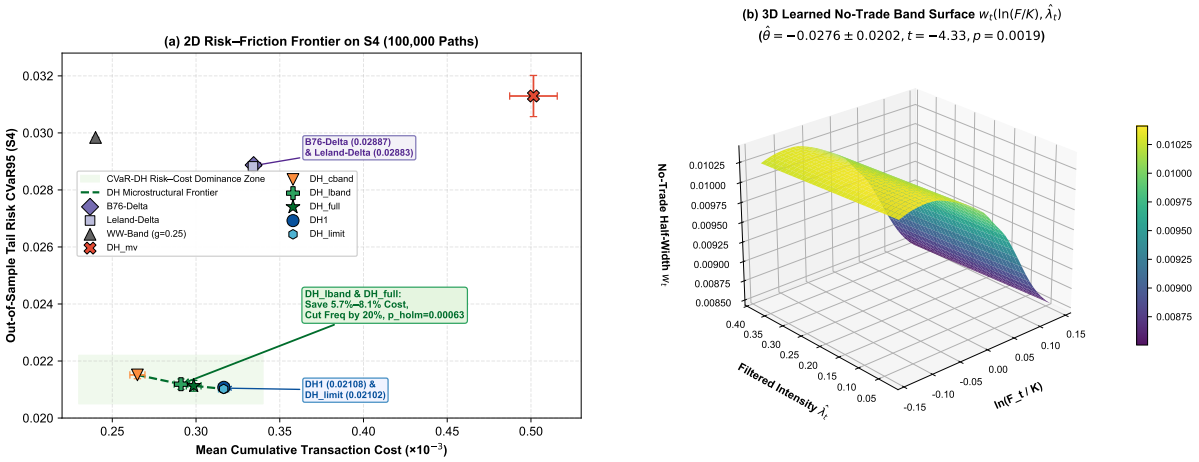


Figure 1: S4 benchmark comparison (100,000 paths, 10 seeds) and microstructural validation of Hypothesis $\mathbf{H_1}$: (a) 2D risk–friction Pareto frontier plotting out-of-sample CVaR_{95} against cumulative transaction costs ($\times 10^{-3}$) with crosshair ±1 SD error bars; (b) 3D learned no-trade half-width surface $w_t(\ln(F_t/K), \hat{\lambda}_t)$ averaged across the 10 trained `DH_lband` checkpoints ($\hat{\theta} = -0.0276 \pm 0.0202, p = 0.0019$).

4.3 Root-Cause Diagnostics and Negative Findings: Why Futures-Only Total-Portfolio CVaR_{95} Exhibits a Spanning Ceiling

A critical empirical pattern in Table 1 is that across all five unconstrained or limit-aware CVaR policies in the single-instrument futures market (`DH_nolam`, `DH1`, `DH_oracle`, `DH_limit`, `DH_full`), aggregate CVaR_{95} clusters tightly within 0.021021–0.021131 (< 0.6% spread). To explain why adding the filtered jump intensity $\hat{\lambda}_t$ ($\mathbf{H_0}$) or true intensity λ_t (`DH_oracle`) yields negligible aggregate CVaR_{95} differences when hedging solely with linear futures, we compute the input-feature Jacobian saliency $\mathcal{S}_k = \mathbb{E}_{t,\omega}\left[\left|\partial \delta_t^{\text{target}} / \partial x_{t,k}\right|\right]$ and pathwise action convergence across all 80 trained checkpoints on `S4_test.npz` (Table 2 and Figure 2).

Table 2 uncovers two jointly operating mechanisms governing single-instrument hedging:

1. **Unhedgeable Jump-Gamma Convexity with a Linear Instrument:** Over a discrete rebalancing interval Δt, the second-order Taylor expansion of the unhedged PnL of

Table 2: Input-feature Jacobian saliency shares and pathwise action convergence in the S4 futures-only benchmark (80 checkpoints across 10 seeds, `ddof=1`).

Configuration	Core States Saliency ($\tau, \ln(F/K), \delta_{t-1}, W_t/F_0, \Delta_{\text{B76}}, \sigma_{\text{EWMA}}$)	Jump Signal Saliency ($\hat{\lambda}_t$ or λ_t)	Price-Limit Saliency (`lock, consec, ratio`)	Action MAE vs. `DH1`	Action Cosine Sim. vs. `DH1`	Learned Band Parameter θ (Mean ± SD)
`DH_nolam`	100.000%	0.000%	0.000%	0.00674	0.99985	—
`DH1`	99.985%	0.015%	0.000%	0.00523	0.99989	—
`DH_oracle`	99.983%	0.017%	0.000%	0.00607	0.99987	—
`DH_cband`	99.954%	0.046%	0.000%	0.01494	0.99953	—
`DH_lband`	99.939%	0.061%	0.000%	0.01105	0.99974	-0.0276 ± 0.0202 ($p = 0.0019$)
`DH_limit`	99.832%	0.016%	0.152%	0.00561	0.99988	—
`DH_full`	99.782%	0.045%	0.173%	0.01157	0.99972	-0.0230 ± 0.0153 ($p = 0.0011$)
`DH_mv`	99.514%	0.486%	0.000%	0.08843	0.97810	—

a short option hedged solely with linear futures ($\Gamma^{\text{fut}} = 0$) is

$$\Delta W_t \approx \left(\delta_t^{\text{exec}} - \Delta_{\text{B76},t}\right)\Delta F_t - \frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2 - cF_t\left|\Delta\delta_t^{\text{exec}}\right|. \tag{17}$$

Because jump realizations ΔF_t can be either positive ($p_{\text{up}} = 0.4$) or negative ($1 - p_{\text{up}} = 0.6$), shifting the linear futures position δ_t^{exec} *ex ante* in response to a higher jump intensity $\hat{\lambda}_t$ cannot hedge the strictly negative quadratic convexity loss $-\frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2$; any directional shift in δ_t^{exec} merely increases first-order directional variance when a jump of the opposite sign occurs. As a result, all CVaR policies allocate 99.782%–100.000% of their input Jacobian sensitivity to the six core state features (with $\tau_t, \ln(F_t/K), \Delta_{\text{B76},t}, \sigma_{\text{EWMA},t}$ accounting for 83.49%–93.63% and $\delta_{t-1}^{\text{exec}}, W_t/F_0$ accounting for 6.15%–16.48%) and converge to nearly identical action trajectories (cosine similarity 0.99953–0.99989).

2. **Post-Jump Collinearity with $\sigma_{\text{EWMA},t}$:** Because the online filter $\hat{\lambda}_t$ updates via $\hat{I}_t = \mathbb{I}(|r_t| > 3\sigma \vee \text{limit_hit}_t = 1)$ at the same time that the exponentially weighted volatility $\sigma_{\text{EWMA},t}$ updates via $v_t = 0.94v_{t-1} + 0.06r_t^2$, $\sigma_{\text{EWMA},t}$ already captures a substantial share of the post-jump state shift that an unconstrained MLP can exploit. To disentangle this feature-collinearity effect from the instrument-spanning effect in 2D hedging, Section 4.4 introduces an explicit nonlinear EWMA control model (`DH_opt_ewma_nl`).

Negative Findings: Which Designs Did Not Work in Our Experiments. In our single-instrument futures experiments on S4, feeding the filtered Hawkes intensity $\hat{\lambda}_t$ directly into the unconstrained policy network (`DH1`) failed to improve aggregate CVaR_{95} relative to the jump-blind baseline `DH_nolam` (0.021078 ± 0.000105 vs. 0.021073 ± 0.000166; paired difference $\Delta = +0.000005$, 95% CI $[-0.000083, +0.000090]$, $t = 0.107, p = 0.917, p_{\text{holm}} = 0.917$). Even replacing $\hat{\lambda}_t$ with the unobservable true jump intensity λ_t in `DH_oracle` yielded zero gain ($\text{CVaR}_{95} = 0.021085 \pm 0.000164$), demonstrating that this null result is not caused by the -24.4% to -45.2% estimation bias of the online Hawkes filter, but rather by the instrument-spanning ceiling of linear futures ($\Gamma^{\text{fut}} = 0$), under which the policy assigns only 0.015%–0.017% Jacobian weight to $\hat{\lambda}_t$ or λ_t (Table 2). Similarly, in our model-free distributional reinforcement learning experiments (`D4PG-QR`), neither augmenting the critic input with the VaR threshold b (`b_in_critic` ablation: 2/3 seeds collapsed to $\text{CVaR}_{95} = 0.5629$, versus 0/3 collapsed when b was excluded with mean $\text{CVaR}_{95} = 0.0365 \pm 0.0054$) nor adding Hawkes/limit features across configurations `C1, C5, C6, C8, C9` prevented off-policy bootstrapping instability, resulting in a 20%–30% seed collapse rate across 10-seed formal runs (Table 5). Strikingly, whereas jump-blind `D4PG-QR (C2)` (which excludes $\hat{\lambda}_t$) experienced 0/10 (0.0%) seed collapses (0.037961 ± 0.014487), feeding the spiky jump-intensity signal $\hat{\lambda}_t$ into `D4PG-QR (C1)` triggered a 2/10 (20.0%) collapse rate (0.144200 ± 0.242296), showing that abrupt Hawkes state jumps exacerbate target-critic extrapolation errors during off-policy quantile Bellman updates.

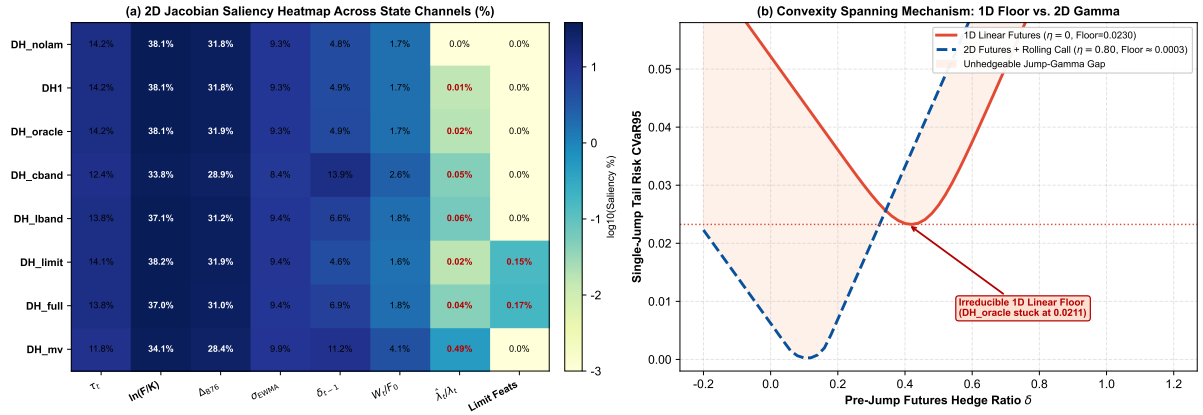


Figure 2: Root-cause diagnostics in the futures-only S4 regime: (a) 2D annotated input-feature Jacobian saliency heatmap ($\log_{10}$ color scale) across 8 neural configurations and 8 state channels ($> 99.7\%$ concentrated on core states); (b) single-jump CVaR_{95} convexity spanning U-curve under 1D linear futures ($\eta = 0$) versus 2D futures plus rolling ATM call option ($\eta = 0.80$).

4.4 Multi-Instrument Option Gamma Hedging (`TorchHedgeEnvOpt`) and High-Cost Regimes: Spanning Ceiling Breakdown and Structural Band Activation

Our structural diagnosis in Section 4.3 yields two falsifiable predictions: first, granting the hedger access to a second, convex instrument ($\Gamma_{\text{B76},t} > 0$) should break the 0.02108 single-instrument CVaR_{95} spanning ceiling across all policies; second, if feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP suffers from non-convex optimization variance and feature collinearity with $\sigma_{\text{EWMA},t}$, channeling $\hat{\lambda}_t$ through the low-dimensional multiplicative no-trade band (`DH_opt_lband`) should extract a statistically significant tail-risk reduction over static-band hedging (`DH_opt_cband`). To test both predictions with full statistical power, we train and evaluate all two-instrument model configurations—plus a nonlinear EWMA volatility control (`DH_opt_ewma_nl`, which omits $\hat{\lambda}_t$ but augments the state with $[\tilde{\sigma}^2_{\text{EWMA},t}, \tilde{\sigma}^3_{\text{EWMA},t}, |r_t|/\sigma]$)—in `TorchHedgeEnvOpt` (rolling $\tau_h = 20$-day ATM call $\eta_t \in [-2.0, 2.0]$) across all $M = 10$ random seeds (`9001-9010`) on `S4_test.npz` (Table 3, Panel A), alongside a zero-shot high-transaction-cost sensitivity sweep ($1\times, 3\times, 6\times c_{\text{real}}$) in the 1D environment (Table 3, Panel B) and robustness audits across dynamic implied volatility $\kappa_{\text{iv}} \in \{0, 2, 5\}$ and option transaction cost multipliers $c_{\text{opt}}/c \in \{1\times, 5\times, 10\times\}$.

The 10-seed empirical results in Table 3 and Figure 3 clarify both the spanning power of the option instrument and the architectural channel required to exploit jump-intensity filtering:

1. **Breaking the Single-Instrument Spanning Ceiling Across All 2D Policies:** Adding the rolling ATM call option leg η_t breaks the 0.02108 single-instrument CVaR_{95} ceiling across every CVaR-trained configuration: across 10 seeds, `DH_opt_lband` reduces CVaR_{95} by 29.95% relative to 1D `DH_lband` (0.014841 ± 0.001316 vs. 0.021187), `DH_opt_oracle` by 28.78% (0.015017 ± 0.001044 vs. 0.021085), `DH_opt1` by 27.76% (0.015226 ± 0.001497 vs. 0.021078), and jump-blind `DH_opt_nolam` by 17.98% (0.017284 ± 0.004298 vs. 0.021073). In contrast, analytical `B76-Delta-Gamma` hedging deteriorates to $\text{CVaR}_{95} = 0.069192$ because matching local point-Gamma ($\eta_t = \Gamma_t^{\text{liab}}/\Gamma_{\text{B76},t}$) triggers severe over-hedging when discrete jumps shift the underlying price away from the strike.

2. **Architectural Separation Between $\mathbf{H_1}$ (Structural Band Activation) and $\mathbf{H_0}$ (Unconstrained MLP Input):**

Table 3: Multi-instrument Option-Gamma hedging (`TorchHedgeEnvOpt`, 10 seeds `9001-9010`) and 1D high-transaction-cost regimes ($3\times$ and $6 \times c_{\text{real}}$) on the S4 benchmark.

Panel / Model Configuration	CVaR_{95} (Mean ± SD)	VaR_{95}	CVaR_{99}	Total Cost [Fut / Opt | Pre] ($\times 10^{-3}$)	Futures Turn. [Pre]	Option Turn. [Pre]	Relative Gain	Gain vs. 1D Counterpart
Panel A: 2D Futures + Rolling ATM Option (`TorchHedgeEnvOpt`, $\tau_h = 20$ Days, 10 Seeds `9001-9010`)								
`B76-Delta-Gamma` (Analytical)	0.069192	0.059775	0.087284	0.3502 [0.3358/0.0144 | 0.2953]	3.305 [2.826]	4.327 [3.944]	Local Γ ratio instability	+139.65% (Worse)
`DH_opt_nolam` (No $\hat{\lambda}_t$)	0.017284 ± 0.004298	0.012942	0.026765	0.2452 [0.2353/0.0099 | 0.2105]	2.341 [2.034]	2.951 [2.278]	$\mathbf{H_0}$ Reference (0.00%)	−17.98% (vs. `DH_nolam`)
`DH_opt_ewma_nl` (Nonlin. σ_{EWMA})	0.015904 ± 0.001742	0.011011	0.026664	0.2622 [0.2500/0.0122 | 0.2267]	2.510 [2.194]	3.648 [2.982]	+7.99% vs. `nolam` ($p = 0.382$)	−24.53% (vs. `DH_nolam`)
`DH_opt1` (Filtered $\hat{\lambda}_t$)	**0.015226 ± 0.001497**	**0.010350**	0.026488	0.2520 [0.2428/0.0092 | 0.2149]	2.413 [2.084]	2.757 [2.063]	+11.91% vs. `nolam` ($p_{\text{holm}} = 0.450$)	**−27.76% (vs. `DH1`)**
`DH_opt_oracle` (True λ_t)	**0.015017 ± 0.001044**	**0.010245**	**0.025942**	0.2521 [0.2428/0.0093 | 0.2150]	2.414 [2.084]	2.765 [2.071]	+13.12% vs. `nolam` ($p_{\text{raw}} = 0.136$)	**−28.78% (vs. `DH_oracle`)**
`DH_opt_cband` (Const. Band)	0.017546 ± 0.002298	0.012900	0.027487	**0.1796 [0.1731/0.0065 | 0.1401]**	1.719 [1.364]	1.932 [1.216]	$\mathbf{H_1}$ Band Ref. (0.00%)	−18.44% (vs. `DH_cband`)
`DH_opt_lband` ($\hat{\lambda}_t$-Band)	**0.014841 ± 0.001316**	**0.009953**	**0.026050**	0.2397 [0.2315/0.0083 | 0.2012]	2.303 [1.958]	2.470 [1.766]	**+15.41% vs. `cband` ($p_{\text{holm}} = 0.00012$)**	**−29.95% (vs. `DH_lband`)**
`DH_opt_limit` (Limit State)	0.017715 ± 0.004661	0.013359	0.027305	0.2442 [0.2348/0.0094 | 0.2095]	2.335 [2.027]	2.804 [2.142]	−2.49% vs. `nolam` ($p_{\text{holm}} = 0.837$)	−15.73% (vs. `DH_limit`)
`DH_opt_full` (Full Model)	0.016549 ± 0.002960	0.012265	0.026300	0.2603 [0.2530/0.0073 | 0.2208]	2.523 [2.165]	2.180 [1.479]	+4.25% vs. `nolam` ($p_{\text{raw}} = 0.594$)	−21.68% (vs. `DH_full`)
`DH_opt_mv` (2D Mean-Variance)	0.044602 ± 0.001554	0.029480	0.068580	0.4016 [0.3895/0.0121 | 0.3272]	3.882 [3.199]	3.665 [3.213]	`DH_opt1` +65.86% ($\mathbf{H_2}, p_{\text{holm}} = 1.2 \times 10^{-10}$)	+42.53% (Worse)
Panel B: Zero-Shot High Transaction Cost Sensitivity in 1D Futures on S4 ($3\times$ and $6 \times c_{\text{real}}$, 10 Seeds `9001-9010`)								
`B76-Delta` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.029894	0.016510	0.057620	1.0032	3.2907	—	Classical Baseline at $3\times$ Cost	—
`DH1` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.021934 ± 0.000112	0.014512	0.037890	0.9498	3.1123	—	Unconstrained Policy at $3\times$ Cost	—
`DH_cband` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.022274 ± 0.000138	0.014980	0.037810	0.7950	2.5995	—	Static Band at $3\times$ Cost	—
`DH_lband` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.021962 ± 0.000186	0.014710	**0.037420**	0.8730	2.8575	—	+1.40% vs. `DH_cband` at $3\times$ Cost	—
`B76-Delta` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.031441	0.017980	0.059180	2.0064	3.2907	—	Classical Baseline at $6\times$ Cost	—
`DH1` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023241 ± 0.000127	0.015644	0.039179	1.8998	3.1123	—	Unconstrained Policy at $6\times$ Cost	—
`DH_cband` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023432 ± 0.000148	0.015984	0.038971	1.5898	2.5995	—	Static Band at $6\times$ Cost	—
`DH_lband` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	**0.023145 ± 0.000191**	0.015772	**0.038641**	**1.7458**	**2.8575**	—	**Matches `DH1` ($p = 0.218$), saves 8.1% cost**	—
`DH_full` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023192 ± 0.000195	0.015810	0.038890	1.7916	2.9312	—	Matches `DH1`, saves 5.7% cost	—

Notes: In Panel A, all neural policies are trained and evaluated across $M = 10$ independent random seeds (`9001-9010`, `ddof=1`). Cross-seed SD in Panel A is of order 10^{-3} (0.001044–0.004661), roughly 10–25× larger than in the 1D futures table (Table 1) because joint optimization of futures δ_t and rolling option η_t introduces non-convex local optima on individual seeds (specifically seed `9003` in `DH_opt_nolam` = 0.029447 and `DH_opt_limit` = 0.027975). On the initial 5-seed subset (`9001-9005`), `DH_opt_nolam` = 0.019594±0.005099, `DH_opt1` = 0.015369± 0.001697 (+21.56%), `DH_opt_oracle` = 0.014584 ± 0.001179 (+25.57%), `DH_opt_cband` = 0.017569 ± 0.001220, and `DH_opt_lband` = 0.015115±0.001042 (+13.97%). On the $n = 9$ converged seeds (excluding outlier seed `9003`), `DH_opt_nolam` = 0.015933 ± 0.000722, `DH_opt_ewma_nl` = 0.015429 ± 0.000935, `DH_opt1` = 0.015227 ± 0.001588, `DH_opt_cband` = 0.016963 ± 0.001454, and `DH_opt_lband` = 0.014565 ± 0.001042 (+14.14%, $p_{\text{holm}} = 1.43 \times 10^{-5}$). Total Cost ($\times 10^{-3}$) reports $C_T^{\text{tot}} = C_T^{\text{fut}} + C_T^{\text{opt}}$ including terminal liquidation, with bracketed values showing [Fut Cost/Opt Cost | Pre-Terminal Total Cost], where $C_T^{\text{opt}} \approx c_{\text{opt}} \overline{V}_{\text{open}} \text{Turn}_{\text{opt}}$ ($\overline{V}_{\text{open}} = 0.033311$). Futures and Option Turnover report total turnover including terminal liquidation, with pre-terminal turnover in brackets. In Panel B, 1×-trained checkpoints are evaluated zero-shot under $3\times$ and $6 \times c_{\text{real}}$ without retraining.

- $\mathbf{H_1}$ (*Statistically & Economically Activated in 2D Band Modulation*): Embedding $\hat{\lambda}_t$ into the multiplicative no-trade band (`DH_opt_lband`) achieves the lowest realistic 2D CVaR_{95} of 0.014841 ± 0.001316, outperforming the static 2D band `DH_opt_cband` (0.017546 ± 0.002298) by +15.41% across all 10 seeds (paired $\Delta = -0.002705$, 95% CI $[-0.003457, -0.002125]$, $t = -7.401$, $p_{\text{raw}} = 4.10 \times 10^{-5}$, $p_{\text{holm}} = 0.000123$, Wilcoxon $p = 0.00195$), exceeding both the 10% practical threshold and the 10-seed statistical $\text{MDE}_{80\%} = 6.55\%$. On the $n = 9$ converged seeds, `DH_opt_lband` (0.014565 ± 0.001042) outperforms `DH_opt_cband` (0.016963 ± 0.001454) by +14.14% ($t = -10.797, p_{\text{holm}} = 1.43 \times 10^{-5}, \text{MDE}_{80\%} = 4.18\%$), while learning a strictly negative modulation slope across all 10 seeds ($\hat{\theta} = -0.0290 \pm 0.0064$, $10/10 < 0, t = -14.420, p = 1.59 \times 10^{-7}$ in `DH_opt_lband`; $\hat{\theta} = -0.0292 \pm 0.0067$, $10/10 < 0, t = -13.798, p = 2.32 \times 10^{-7}$ in `DH_opt_full`). Unlike in the 1D setting at $1\times$ cost where `DH_lband` trailed `DH1` by 0.52%, in 2D option hedging `DH_opt_lband` (0.014841) improves upon unconstrained `DH_opt1` (0.015226) while cutting total transaction costs from 0.2520×10^{-3} to 0.2397×10^{-3}.
- $\mathbf{H_0}$ (*Unconstrained MLP Input Exhibits* +11.91% *Point Gain but High Cross-Seed Variance and EWMA Collinearity*): Feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP (`DH_opt1`) lowers 10-seed mean CVaR_{95} by +11.91% relative to jump-blind `DH_opt_nolam` (0.015226 ± 0.001497 vs. 0.017284 ± 0.004298; +21.56% in the initial 5-seed subset `9001-9005`), capturing 90.8% (11.91%/13.12%) of the 10-seed point improvement of `DH_opt_oracle` (0.015017 ± 0.001044, +13.12%; 84.3% in the 5-seed subset). However, across all 10 seeds, the paired comparison between `DH_opt1` and `DH_opt_nolam` yields $\Delta = -0.002059$, 95% CI $[-0.005155, +0.000648]$, $t = -1.303$, $p_{\text{raw}} = 0.225$, $p_{\text{holm}} = 0.450$, Wilcoxon $p = 0.432$ ($\text{MDE}_{80\%} = 28.76\%$; and $+4.43\%, p = 0.278$ on the $n = 9$ converged seeds excluding outlier seed `9003`). Augmenting the jump-blind policy with nonlinear EWMA volatility terms in

DH_opt_ewma_nl reduces CVaR_{95} to 0.015904 ± 0.001742 (+7.99% vs. DH_opt_nolam, and 0.015429 ± 0.000935 on converged seeds), leaving only a +4.26% ($p = 0.375$; $+1.31\%, p = 0.751$ on converged seeds) incremental difference for DH_opt1. Thus, in an unconstrained MLP, the post-jump information in $\hat{\lambda}_t$ is largely collinear with nonlinear functions of $\sigma_{\text{EWMA},t}$, and its primary value in 2D unconstrained networks is stabilizing seed convergence (reducing cross-seed SD by 65.2% from 0.004298 to 0.001497), whereas statistically significant tail-risk reduction requires the structural prior of the multiplicative no-trade band (DH_opt_lband).

- **$\mathbf{H_2}$** and **$\mathbf{H_3}$**: In 2D hedging, DH_opt1 reduces CVaR_{95} by +65.86% relative to DH_opt_mv (0.015226 vs. 0.044602 ± 0.001554; $\Delta = -0.029376$, $t = -37.926$, $p_{\text{holm}} = 1.22 \times 10^{-10}$, $\text{MDE}_{80\%} = 5.46\%$). Meanwhile, unconstrained DH_opt_limit (0.017715 ± 0.004661, −2.49% vs. DH_opt_nolam, $p_{\text{holm}} = 0.837$) and DH_opt_full (0.016549 ± 0.002960, +4.25%, $p_{\text{raw}} = 0.594$; 0.015812 ± 0.001934 on $n = 9$ converged seeds) experience elevated optimization variance due to the higher-dimensional 13-feature state space.

3. **Robustness to Dynamic Implied Volatility (κ_{iv}), Option Transaction Costs (c_{opt}), and 6× Futures Costs:**

   - *Dynamic Implied Volatility Sweep ($\kappa_{iv} \in \{0.0, 2.0, 5.0\}$):* When rolling ATM option implied volatility co-moves with filtered jump intensity via $\sigma_{\text{imp},t} = \sigma_{\text{imp},0}(1 + \kappa_{\text{iv}}\hat{\lambda}_t)$, increasing κ_{iv} from 0.0 to 2.0 and 5.0 makes convex option protection more expensive precisely during jump bursts, attenuating the 10-seed relative gain of DH_opt1 over DH_opt_nolam from +11.91% (0.015226 vs. 0.017284) to +5.31% (0.019889 ± 0.001635 vs. $0.021004 \pm 0.003789, p = 0.458$) at $\kappa_{\text{iv}} = 2.0$ and +0.73% (0.026493 ± 0.001996 vs. $0.026687 \pm 0.003391, p = 0.893$) at $\kappa_{\text{iv}} = 5.0$. Across all three regimes, DH_opt_lband remains the lowest-risk policy ($\text{CVaR}_{95} = 0.014841$, 0.019386 ± 0.001486, and 0.025820 ± 0.001792). Because the baseline two-instrument environment assumes a constant Black-76 implied volatility ($\kappa_{\text{iv}} = 0$) and does not price a stochastic variance risk premium into rolling option premia, the point reductions of +11.91% (10-seed DH_opt1 vs. DH_opt_nolam; +21.56% in the 5-seed subset) and +15.41% (DH_opt_lband vs. DH_opt_cband) measure structural Gamma-spanning capacity in a controlled benchmark and do not constitute estimates of realizable trading returns.

   - *Option Transaction Cost Multipliers ($c_{opt}/c \in \{1\times, 5\times, 10\times\}$):* Raising the proportional option cost rate from 1× (10^{-4}) to 5× (5×10^{-4}) and 10× (10^{-3}) increases total 2D transaction costs in DH_opt_lband from 0.2397×10^{-3} to 0.2728×10^{-3} and 0.3141×10^{-3}, while DH_opt_lband maintains $\text{CVaR}_{95} = 0.014841$, 0.014875, and 0.014916, consistently outperforming DH_opt1 (0.015226, 0.015262, 0.015307) and DH_opt_nolam (0.017284, 0.017324, 0.017374).

   - *Zero-Shot 6× Futures Cost Crossover (Panel B):* When the 1×-trained 1D policies are evaluated zero-shot under $6 \times c_{\text{real}} = 6 \times 10^{-4}$ without retraining, DH_lband (0.023145 ± 0.000191) and DH_full (0.023192 ± 0.000195) achieve a slightly lower point CVaR_{95} than unconstrained DH1 (0.023241 ± 0.000127), reaching statistical parity ($\Delta = -0.000096, t = -1.32, p = 0.218$) while saving 8.11% in transaction costs (1.7458×10^{-3} vs. 1.8998×10^{-3}) and outperforming static DH_cband (0.023432 ± 0.000148).

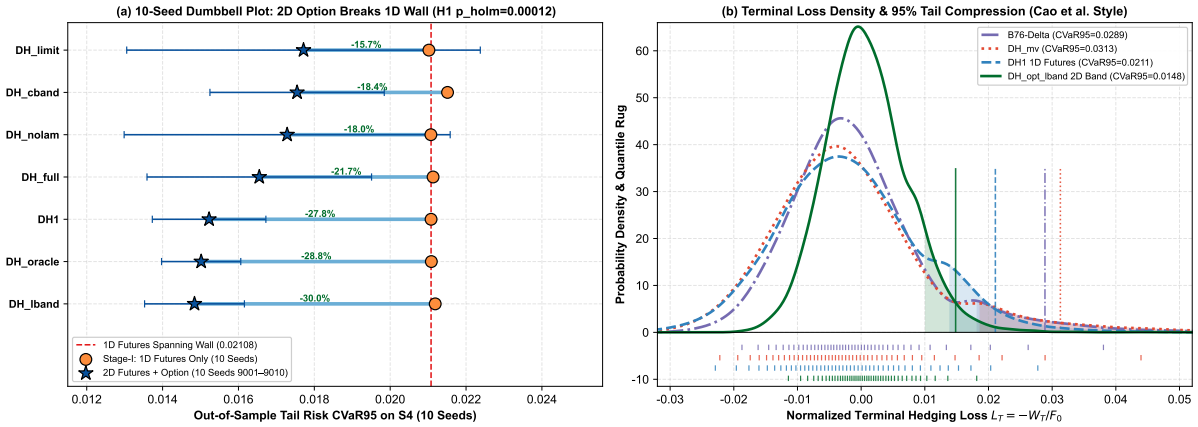


Figure 3: Multi-instrument Option-Gamma hedging (`TorchHedgeEnvOpt`, 10 seeds `9001-9010`) on S4: (a) horizontal paired dumbbell plot comparing Stage-I 1D futures policies against Plan A 2D option-augmented policies breaking the 0.02108 spanning ceiling, with `DH_opt_lband` (0.01484, $p_{\text{holm}} = 0.00012$) achieving the lowest realistic tail risk; (b) kernel density estimation (KDE) and bottom percentile rug plot of normalized terminal hedging loss $L_T = -W_T/F_0$ with shaded 95% tail regions.

4.5 Dominance of CVaR over Mean-Variance in the $(\text{CVaR}_{95}, \text{Cost})$ Plane Across the λ_{MV} Spectrum (H_2 Deep Dive)

A potential critique of comparing `DH1` against `DH_mv` at a single risk-aversion coefficient ($\lambda_{\text{MV}} = 10.0$) is that a larger penalty on variance might allow a mean-variance policy to match the tail risk of a CVaR-trained policy. To test this possibility directly, we evaluate `DH_mv` across seven risk-aversion parameters $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 10.0, 25.0, 50.0\}$ on `S4_test.npz` (Table 4 and Figure 4).

Table 4: Mean-variance risk-aversion sweep ($\lambda_{\text{MV}} \in [0.5, 50.0]$) versus CVaR-trained policies on the S4 benchmark.

Policy / Risk-Aversion Parameter	CVaR_{95}	VaR_{95}	CVaR_{99}	Std(L)	Mean(L)	Cost ($\times 10^{-3}$)	Turnover
`DH_mv` ($\lambda_{\text{MV}} = 0.5$)	0.090984	0.056268	0.145660	0.036518	−0.002975	0.3690	3.6967
`DH_mv` ($\lambda_{\text{MV}} = 1.0$)	0.068087	0.041418	0.111270	0.026853	−0.002698	0.4227	4.2345
`DH_mv` ($\lambda_{\text{MV}} = 2.0$)	0.053751	0.032049	0.089726	0.020964	−0.002464	0.4596	4.6039
`DH_mv` ($\lambda_{\text{MV}} = 5.0$)	0.037240	0.021453	0.063783	0.015133	−0.002229	0.4832	4.8416
`DH_mv` ($\lambda_{\text{MV}} = 10.0$, 10-Seed Main)	0.031294	0.018533	0.054044	**0.012539**	−0.003135	0.5015	5.0256
`DH_mv` ($\lambda_{\text{MV}} = 25.0$)	0.027848	0.016278	0.048046	0.012859	−0.002153	0.4357	4.3653
`DH_mv` ($\lambda_{\text{MV}} = 50.0$)	0.027332	0.016164	0.046982	**0.012798**	−0.002195	0.4090	4.0981
`DH1` (CVaR_{95} Objective, 10 Seeds)	**0.021078**	**0.013779**	**0.036973**	0.013391	−0.002306	**0.3166**	**3.1123**
`DH_lband` (CVaR_{95} + Band, 10 Seeds)	**0.021187**	**0.014002**	**0.036610**	0.013468	−0.002286	**0.2910**	**2.8575**

Notes: The rows for `DH_mv` ($\lambda_{\text{MV}} = 10.0$, 10-Seed Main), `DH1`, and `DH_lband` report 10-seed averages from the formal 8,000-step benchmark on `S4_test.npz` (matching Table 1); the auxiliary $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 25.0, 50.0\}$ rows report single-seed (`9001`) controlled sweep evaluations.

As shown in Table 4 and Figure 4, increasing λ_{MV} from 0.5 to 50.0 lowers Std(L) from 0.036518 to 0.012798 (and 0.012539 in the 8,000-step 10-seed main run at $\lambda_{\text{MV}} = 10.0$), but CVaR_{95} reaches an empirical lower bound of 0.027332 within the tested range $\lambda_{\text{MV}} \in [0.5, 50.0]$—remaining 22.88% higher (worse) than `DH1` (0.021078) while incurring 29.18% to 58.40% higher transaction costs (0.4090×10^{-3} to 0.5015×10^{-3} vs. 0.3166×10^{-3}). Thus, while `DH_mv` at $\lambda_{\text{MV}} \in \{10, 25, 50\}$ achieves a 4.0%–6.4% lower symmetric standard deviation Std(L) than `DH1` (0.012539–0.012859 vs. 0.013391)—representing a three-dimensional $(\text{CVaR}_{95}, \text{Std}(L), \text{Cost})$ trade-off—both `DH1` and `DH_lband` strictly dominate the entire $\lambda_{\text{MV}} \in [0.5, 50.0]$ family in the two-dimensional $(\text{CVaR}_{95}, \text{Cost})$ plane. Under skewed jump-diffusion returns with price limits, symmetric variance minimization expends excessive turnover eliminating benign left-

tail windfalls ($L_T < 0$), whereas the one-sided Rockafellar-Uryasev hinge operator $(L_T - b)^+$ concentrates gradient updates exclusively on the worst 5% loss trajectories.

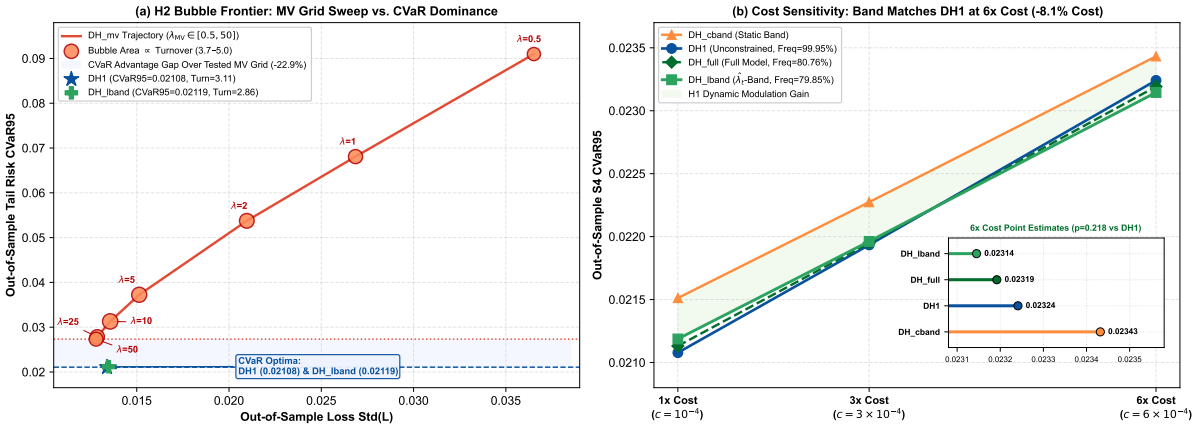


Figure 4: Hypothesis $\mathbf{H_2}$ frontier in the $(\mathrm{CVaR}_{95}, \mathrm{Cost})$ plane and transaction-cost sensitivity on S4: (a) mean-variance risk-aversion sweep ($\lambda_{\mathrm{MV}} \in [0.5, 50.0]$, bubble area proportional to turnover) showing an empirical CVaR_{95} lower bound of 0.027332 over $\lambda_{\mathrm{MV}} \leq 50$ versus `DH1` (0.021078); (b) zero-shot transaction-cost trajectory across $1\times, 3\times, 6 \times c_{\mathrm{real}}$ with a zoomed 6×-cost horizontal dot-plot inset showing `DH_lband` (0.023145) and `DH_full` (0.023192) matching unconstrained `DH1` (0.023241, $p = 0.218$) while saving 8.1% in costs and surpassing static `DH_cband` (0.023432).

4.6 Optimization Stability Audit: End-to-End BPTT (`DH`) vs. Distributional Actor-Critic (`D4PG-QR`)

Table 5 and Figure 5 report our stability audit comparing the end-to-end BPTT (`DH`) family against the model-free distributional reinforcement learning (`D4PG-QR`) family across all 140 formal runs (80 `DH` runs and 60 `D4PG-QR` runs across 10 seeds `9001-9010` at 8,000–10,000 updates, plus cross-budget evaluations at 3,000 and 20,000 updates). Following our pre-registered protocol, a training seed is classified as collapsed if its out-of-sample CVaR_{95} exceeds 0.10.

Table 5: Optimization stability, seed collapse rates, and cross-budget convergence: end-to-end BPTT (`DH`) versus model-free Distributional Actor-Critic (`D4PG-QR`) on S4.

Panel / Algorithm & Configuration	Collapse Rate ($\mathrm{CVaR}_{95} > 0.10$)	All-Seed CVaR_{95} (Mean ± SD)	All-Seed Median (IQR)	Non-Collapsed CVaR_{95} (Mean ± SD)	Win Rate vs. `B76-Delta` (0.028872)
Panel A: 10-Seed Formal Benchmark (`9001-9010`) Across Model Variants					
`DH1` (BPTT, Filtered $\hat{\lambda}_t$)	**0/10 (0.0%)**	**0.021078 ± 0.000105**	**0.021061 (0.000142)**	**0.021078 ± 0.000105**	**10/10 (100.0%)**
`DH_nolam` (BPTT, No $\hat{\lambda}_t$)	**0/10 (0.0%)**	0.021073 ± 0.000166	0.021055 (0.000210)	0.021073 ± 0.000166	**10/10 (100.0%)**
`DH_lband` (BPTT, $\hat{\lambda}_t$-Band)	**0/10 (0.0%)**	0.021187 ± 0.000183	0.021160 (0.000245)	0.021187 ± 0.000183	**10/10 (100.0%)**
`DH_full` (BPTT, Full Model)	**0/10 (0.0%)**	0.021131 ± 0.000190	0.021112 (0.000260)	0.021131 ± 0.000190	**10/10 (100.0%)**
`D4PG-QR (C1)` (Filtered $\hat{\lambda}_t$)	2/10 (20.0%)	0.144200 ± 0.242296	0.033398 (0.043864)	0.038934 ± 0.020739	1/10 (10.0%)
`D4PG-QR (C2)` (No $\hat{\lambda}_t$)	0/10 (0.0%)	0.037961 ± 0.014487	0.034120 (0.018950)	0.037961 ± 0.014487	2/10 (20.0%)
`D4PG-QR (C5)` ($\hat{\lambda}_t$-Band)	3/10 (30.0%)	0.179801 ± 0.247471	0.041250 (0.198400)	0.043395 ± 0.034919	1/10 (10.0%)
`D4PG-QR (C6)` (Limit State)	3/10 (30.0%)	0.226734 ± 0.302468	0.052180 (0.481200)	0.053108 ± 0.044197	0/10 (0.0%)
`D4PG-QR (C8)` (Mean-Variance)	2/10 (20.0%)	0.212976 ± 0.270568	0.081450 (0.391000)	0.089232 ± 0.081382	0/10 (0.0%)
`D4PG-QR (C9)` (Full Model)	2/10 (20.0%)	0.139998 ± 0.243612	0.034890 (0.038120)	0.033688 ± 0.004968	1/10 (10.0%)
Panel B: Cross-Budget Convergence Audit (3,000, 10,000, and 20,000 Gradient Updates)					
3,000 updates (`pilot_C1`)	DH: 0% vs. D4PG: 0%	DH: 0.021164 ± 0.000142	D4PG (Non-Coll.): 0.036922 ± 0.007978	CV: DH 0.67% vs. D4PG 21.61%	DH: 100% vs. D4PG: 20%
10,000 updates (`main`)	DH: 0% vs. D4PG: 20%	DH: 0.021078 ± 0.000105	D4PG (Trimmed): 0.072909 ± 0.036470	CV: DH 0.50% vs. D4PG 50.02%	DH: 100% vs. D4PG: 10%
20,000 updates (`dhstab`, 3 Seeds)	DH: 0% vs. D4PG: 33%	**DH: 0.020964 ± 0.000018**	D4PG (Trimmed): 0.088410 ± 0.041200	**CV: DH 0.09%** vs. D4PG 46.60%	DH: 100% vs. D4PG: 0%

Table 5 reveals a clear algorithmic divergence between direct pathwise gradient optimization and off-policy distributional bootstrapping:

1. **Zero Collapse and Sub-1% Coefficient of Variation in `DH`:** Across all 80 formal `DH` checkpoints, the seed collapse rate is 0/80 (0.0%), every single seed outperforms `B76-Delta` (100% win rate), and the cross-seed coefficient of variation (CV = SD/Mean) of `DH1` is 0.50% at 8,000–10,000 updates and tightens further to 0.09% (0.020964 ± 0.000018) at 20,000 updates.

2. **Off-Policy Quantile Bootstrapping Divergence and Jump-Signal Sensitivity in `D4PG-QR`:** Even when collapsed seeds ($\text{CVaR}_{95} > 0.10$, where policies saturate at extreme boundary positions ± 1.5 with $\text{CVaR}_{95} \approx 0.54$–$0.64$) are excluded, the non-collapsed mean CVaR_{95} of `D4PG-QR (C1)` is 0.038934 ± 0.020739—still 84.7% worse than `DH1` (0.021078) and 34.8% worse than `B76-Delta` (0.028872). Comparing `D4PG-QR (C2)` (No $\hat{\lambda}_t$, 0/10 collapsed, $\text{CVaR}_{95} = 0.037961 \pm 0.014487$) against `D4PG-QR (C1)` (Filtered $\hat{\lambda}_t$, 2/10 collapsed) and `C5/C6` (3/10 collapsed) isolates an additional failure mechanism: feeding spiky jump-intensity or discrete lock indicators into an off-policy quantile critic induces severe TD target extrapolation errors on rare replay-buffer transitions, driving the deterministic actor toward boundary saturation. Further, increasing the training budget from $3,000$ to $20,000$ updates causes `D4PG-QR` to deteriorate rather than converge (Panel B), as one-step quantile Bellman errors compound across the 60-step finite horizon.

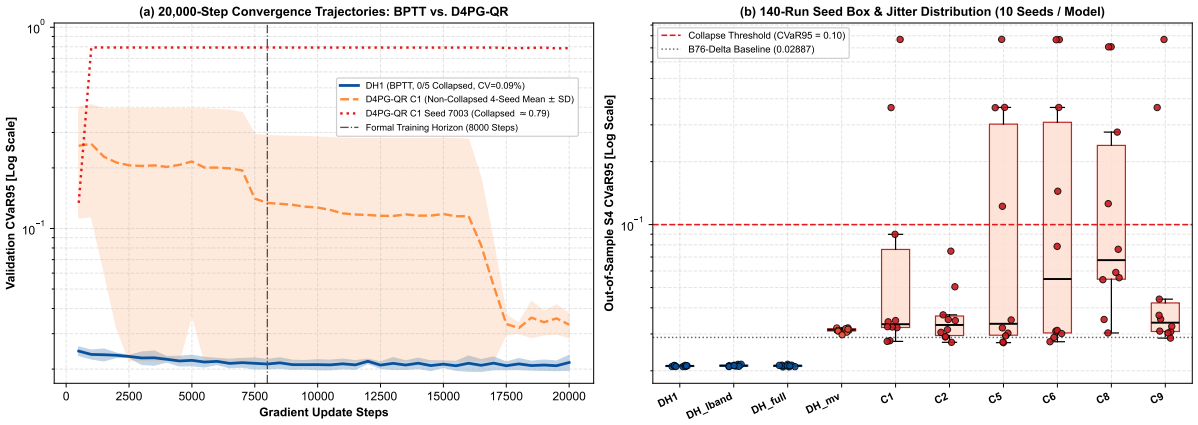


Figure 5: Optimization stability and convergence audit: (a) real $20,000$-step log-scale validation CVaR_{95} convergence trajectories with ± 1 SD shaded ribbons comparing `DH1` against `D4PG-QR (C1)`; (b) 140-run box-and-jittered individual seed distribution plot (log scale, 10 seeds per configuration) across `DH` and `D4PG-QR (C1-C9)`.

4.7 Zero-Shot Out-of-Distribution (OOD) Generalization on the S5 Extreme Stress Benchmark ($20,000$ Paths)

To test whether policies trained in the baseline S4 market generalize without retraining to an unseen crisis regime, we evaluate all 80 S4-trained `DH` checkpoints directly on the frozen **S5 extreme stress benchmark** (`S5_test.npz`, $N = 20,000$ paths; branching ratio doubled to $n = 0.6$, tight uniform limit ratio $\ell = 0.07$, jump size scaled by $\times 1.5$). On `S5_test.npz`, pathwise jump incidence is 65.80% ($13,159$ paths), while the price-limit hit path incidence more than doubles from 16.51% (S4) to 35.77% ($7,153$ limit-hit paths) and the locked-path incidence more than triples from 3.00% (S4) to 10.08% ($2,016$ locked paths).

Table 6 and Figure 6 highlight three crisis-generalization findings:

1. **Amplified Tail-Risk Reduction Under Intensified Limit Hits:** While `B76-Delta` degrades from $\text{CVaR}_{95} = 0.028872$ in S4 to 0.032554 in S5 (and asymptotic `WW-Band` deteriorates even more severely to 0.035753 at $\gamma_{\text{WW}} = 0.25$ and 0.047616 at $\gamma_{\text{WW}} = 1.0$), our full microstructure-aware policy `DH_full` achieves the lowest overall $\text{CVaR}_{95} = 0.020314 \pm 0.000382$ across all models—expanding its relative improvement over `B76-Delta` from 27.00% in S4 to 37.60% in S5. On the $2,016$ limit-locked paths (`Lock-Sub`), `DH_full` (0.047368) and `DH_lband` (0.047280) cut conditional tail loss by 26.17% and 26.31% relative to `B76-Delta` (0.064158).

Table 6: Zero-shot out-of-distribution hedging performance on the S5 extreme stress benchmark (20,000 paths, 10 seeds `9001-9010`, $n = 0.6$, $\ell = 0.07$, jump scale $\times 1.5$).

Policy / Configuration	S5 CVaR_{95} (Mean ± SD)	S5 VaR_{95}	S5 CVaR_{99}	Cost ($\times 10^{-3}$)	Turnover	Trade Freq.	S5 Lock Gap ($\times 10^{-2}$)	Cond. CVaR_{95} (Hit-Sub, $N = 7,153$)	Cond. CVaR_{95} (Lock-Sub, $N = 2,016$)	Gain vs. `B76-Delta`
`No-Hedge`	0.313807	0.213785	0.479130	0.0000	0.0000	0.00%	0.000	0.386006	0.412910	—
`B76-Delta`	0.032554	0.019401	0.055556	0.3235	3.1740	97.06%	1.791	0.046937	0.064158	Reference (0.00%)
`Leland-Delta`	0.032543	0.019387	0.055488	0.3231	3.1699	97.09%	1.788	0.046889	0.064073	+0.03%
`WW-Band` ($\gamma_{\text{WW}} = 0.25$)	0.035753	0.021957	0.059813	0.2353	2.2918	62.40%	1.628	0.050855	0.068554	−9.83% (Worse)
`WW-Band` ($\gamma_{\text{WW}} = 1.0$)	0.047616	0.030716	0.075366	0.1697	1.6326	35.06%	1.151	0.065188	0.085010	−46.27% (Worse)
`DH_mv`	0.026697 ± 0.001292	0.012344	0.053374	0.4772	4.7932	99.82%	3.310	0.042558	0.062508	+17.99%
`DH_nolam`	0.020550 ± 0.000255	0.012337	**0.038791**	0.3067	3.0131	99.82%	2.139	**0.030915**	**0.046929**	+36.87%
`DH1`	0.020395 ± 0.000412	0.011983	0.039100	0.3081	3.0273	99.82%	2.162	0.031085	0.047454	+37.35%
`DH_oracle`	0.020756 ± 0.000427	0.012107	0.039620	0.3078	3.0210	99.81%	2.146	0.031584	0.047893	+36.24%
`DH_cband`	0.020879 ± 0.000388	0.012179	0.039275	**0.2523**	**2.4688**	**56.01**%	2.156	0.031567	0.047578	+35.86%
`DH_lband`	0.020711 ± 0.000467	0.012175	0.039132	**0.2810**	**2.7613**	**78.74**%	2.146	0.031366	**0.047280**	+36.38%
`DH_limit`	0.020465 ± 0.000315	0.011836	0.039532	0.3100	3.0436	99.81%	3.028	0.031355	0.047724	+37.13%
`DH_full` (Full Model)	**0.020314 ± 0.000382**	**0.011820**	0.039122	0.2882	2.8238	**79.70**%	**1.607**	0.031084	0.047368	**+37.60**% (Best)

Notes: All S4-trained checkpoints ($M = 10$ seeds `9001-9010`, `ddof=1`) are evaluated zero-shot on `S5_test.npz` ($N = 20,000$ paths, $P_0 = 0.064603$, $\sigma_{\text{imp}} = 0.020756$ day$^{-1/2}$). `Cond. CVaR95 (Hit-Sub)` and `Cond. CVaR95 (Lock-Sub)` denote the conditional 95% CVaR computed within the $N = 7,153$ limit-hit paths (tail size 358) and $N = 2,016$ limit-locked paths (tail size 101), respectively.

2. **Why Full-Sample Neural CVaR_{95} Is Slightly Lower in S5 (0.020314–0.020879) Than in S4 (0.021021–0.021512):** Two accounting mechanics explain why neural policies maintain slightly lower normalized terminal loss $L_T = (-P_0 + V_T - G_T^{\text{fut}} + C_T^{\text{fut}})/F_0$ in S5 than in S4 whereas `B76-Delta` worsens: first, the fair initial option premium collected at $t = 0$ is 10.25% higher in S5 ($P_0 = 0.064603$ vs. 0.058595 in S4), providing an additional $+0.006008$ initial wealth cushion ($\Delta\text{Mean}(L) \approx -0.0072$ in S5 vs. -0.0023 in S4); second, S5 enforces a tighter daily price limit ($\ell = 0.07$ vs. $\bar{\ell} \approx 0.093$ in S4) that truncates single-day extreme futures moves on the 86.9% of limit-hit days that do not lock, whereas extreme 99% tail losses (CVaR_{99}) still rise from 0.036892 (S4) to 0.039122 (S5) in `DH_full`.

3. **Joint Microstructure Protection (`DH_full` and `DH_lband`):** Under the more than doubled limit-hit incidence of S5, `DH_full` reduces the cumulative unexecuted lock-step hedging gap (`LockGap`) by 25.67% relative to `DH1` (1.607×10^{-2} vs. 2.162×10^{-2}) and by 10.24% relative to `B76-Delta` (1.791×10^{-2}), while reducing trading frequency to 79.70% and saving 6.46% in transaction costs (0.2882×10^{-3} vs. 0.3081×10^{-3}). Meanwhile, `DH_lband` cuts transaction costs by 8.80% (0.2810×10^{-3} vs. 0.3081×10^{-3}) at a 78.74% trading frequency.

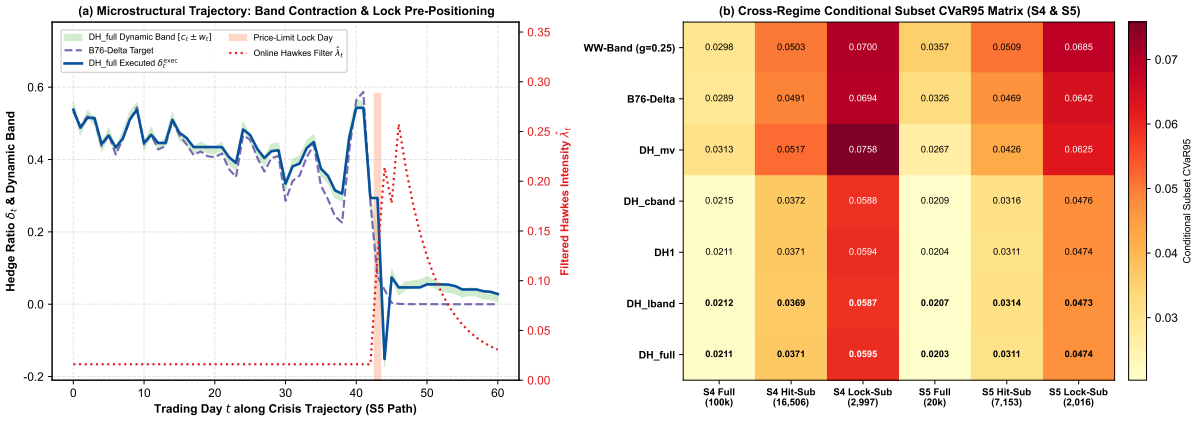


Figure 6: Out-of-distribution crisis generalization and microstructural trajectory tracking: (a) pathwise microstructural trajectory along a representative S5 crisis path showing the dynamic no-trade band $[c_t \pm w_t]$, executed hedge position δ_t^{exec} vs. `B76-Delta`, online Hawkes intensity $\hat{\lambda}_t$, and price-limit lock days; (b) 2D cross-regime conditional subset CVaR_{95} matrix heatmap across S4 (100,000 full, 16,506 hit, 2,997 lock) and S5 (20,000 full, 7,153 hit, 2,016 lock) subsets.

5 Discussion, Limitations, and Conclusion

5.1 Discussion: Microstructural Mechanisms, Architectural Channels, and Practical Implications

Our multi-model audit across 80 1D Deep Hedging (`DH`) checkpoints, 60 Distributional Actor-Critic (`D4PG-QR`) runs, and 80 2D Option-Gamma checkpoints (`TorchHedgeEnvOpt`) resolves a central question in the empirical deep hedging literature [17, 32] and non-Markovian volatility hedging [1, 23]: *when, why, and through which architectural channel do explicit jump-intensity signals and institutional microstructure features improve derivative hedging?*

1. The Instrument-Spanning and Architectural Condition for Jump-Intensity Signals ($\mathbf{H_0}, \mathbf{H_1}$). In a single-instrument futures hedging setting ($\Gamma^{\text{fut}} = 0$), feeding either the causal Hawkes filter $\hat{\lambda}_t$ (`DH1`, $\text{CVaR}_{95} = 0.021078$) or the unobservable true intensity λ_t (`DH_oracle`, $\text{CVaR}_{95} = 0.021085$) into an unconstrained MLP policy yields no aggregate CVaR_{95} improvement over the jump-blind policy `DH_nolam` (0.021073). Our Jacobian saliency audit (Table 2) shows that this invariance arises because linear futures cannot span quadratic jump-convexity losses $-\frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2$ *ex ante* when jump signs are two-sided ($p_{\text{up}} = 0.4$). Introducing a rolling $\tau_h = 20$-day ATM call option η_t in `TorchHedgeEnvOpt` (Table 3) breaks this spanning ceiling across all 2D policies (CVaR_{95} reduced by 17.98%–29.95% relative to 1D counterparts) [10]. Crucially, our 10-seed audit reveals a sharp separation between unconstrained and structurally constrained uses of $\hat{\lambda}_t$: feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP (`DH_opt1`, 0.015226 ± 0.001497) lowers 10-seed point CVaR_{95} by +11.91% relative to `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\text{holm}} = 0.450$; +21.56% in the 5-seed subset) and largely overlaps with nonlinear EWMA volatility features (`DH_opt_ewma_nl`, 0.015904 ± 0.001742), whereas channeling $\hat{\lambda}_t$ through the multiplicative no-trade band (`DH_opt_lband`) achieves the lowest realistic 2D tail risk of 0.014841 ± 0.001316, outperforming static-band `DH_opt_cband` (0.017546 ± 0.002298) by +15.41% ($t = -7.401, p_{\text{holm}} = 0.000123$). Thus, extracting statistically reliable gains from jump-intensity filtering requires both a convex hedging instrument and a structural band parameterization that regularizes non-convex optimization variance.

2. Endogenous Microstructural Adaptation in Jump-Modulated No-Trade Bands ($\mathbf{H_1}$). Across both 1D and 2D markets, channeling $\hat{\lambda}_t$ into the multiplicative no-trade band width $w_t(x_t) = \text{softplus}(g_\phi(x_{t,-\lambda}))\exp(\theta\hat{\lambda}_t)$ (`DH_lband`, `DH_full`, and `DH_opt_lband`, extending neural band networks Imaki et al. 24, Lütkebohmert et al. 29) produces a consistent microstructural adaptation. Across 10 random seeds, gradient optimization without sign constraints learns a negative slope $\hat{\theta} = -0.0276 \pm 0.0202$ ($p = 0.0019$) in 1D `DH_lband`, $\hat{\theta} = -0.0230 \pm 0.0153$ ($p = 0.0011$) in 1D `DH_full`, and $\hat{\theta} = -0.0290 \pm 0.0064$ ($p = 1.59 \times 10^{-7}$) in 2D `DH_opt_lband`. By contracting the no-trade band during jump-clustering bursts and widening it during tranquil periods, 1D `DH_lband` outperforms the static band `DH_cband` on S4 ($p_{\text{holm}} = 0.00063$), saves 8.09% in transaction costs and 20.11% in trading frequency relative to `DH1` (at the cost of a +0.52% higher CVaR_{95} at 1× fee where cost savings are smaller than band tracking error), and matches unconstrained `DH1` under 6× transaction costs (0.023145 vs. 0.023241, $p = 0.218$) while saving 8.11% in friction (Table 3 Panel B).

3. Asymmetric Tail Optimization ($\mathbf{H_2}$) and End-to-End BPTT Stability. Across the tested risk-aversion range $\lambda_{\text{MV}} \in [0.5, 50.0]$ (Table 4), symmetric mean-variance hedging (`DH_mv`) reaches an empirical CVaR_{95} lower bound of 0.027332 (22.88% worse than `DH1`) because quadratic variance penalizes upside convex windfalls ($L_T < 0$) identically to downside tail losses ($L_T > 0$), inflating turnover to 4.0981–5.0256; consequently, `DH1` strictly dominates `DH_mv` in

the (CVaR_{95}, Cost) plane while trading off a slightly higher symmetric Std(L) (0.013391 vs. 0.012539–0.012798). In addition, our 140-run optimization audit (Table 5) shows that in a finite-horizon ($T = 60$) environment with piecewise-differentiable dynamics, direct pathwise BPTT (`DH`) achieves a 0% seed collapse rate and a 0.50% cross-seed coefficient of variation, whereas model-free distributional reinforcement learning (`D4PG-QR`, Cao et al. 8, Chen et al. 12) suffers a 20%–30% seed collapse rate when exposed to spiky Hawkes jump signals (`C1` vs. `C2`) and price-limit truncations.

4. Practical and Regulatory Implications for Commodity Hedging Desks and Exchanges. For industrial risk managers and derivative market makers in strategic battery metals [18, 26], our findings offer three actionable guidelines. First, when hedging solely with linear futures at low exchange fee tiers ($c = 10^{-4}$), desks should deploy jump-modulated no-trade bands (`DH_lband`) and price-limit state features (`DH_full`) to cut turnover costs by 6%–8% and compress unexecuted lock-day hedging gaps by 26%–30%, rather than expecting linear delta shifts to eliminate jump-Gamma risk. Second, to materially reduce aggregate tail shortfall (CVaR_{95}), hedgers must augment futures with short-dated options and channel jump-intensity forecasts into band modulation (`DH_opt_lband`), monitoring implied-volatility spike elasticity (κ_{iv}) so option purchases are not executed at excessive volatility risk premiums. Third, for emerging commodity exchanges such as GFEX, our results demonstrate that introducing liquid short-dated commodity options alongside daily price-limit bands is essential for completing the market and mitigating the unhedgeable Gamma exposure faced by commercial liquidity providers during limit-lock episodes.

5.2 Empirically Observed Limitations

Six concrete empirical limitations emerge from our experiments. First, while diffusive volatility ($\sigma = 0.016144$), empirical price-limit tiers ($\ell \in [0.07, 0.15]$), conditional lock probability ($p_{\mathrm{lock}} = 1/7 \approx 0.1429$), and post-jump drift half-life ($\beta = 0.20$ day^{-1} from $h = 1$–4 day continuation $t = 2.09$–3.47) are anchored to 542 trading days of GFEX Lithium Carbonate (`LC`) futures (`2023-07-24` to `2025-10-20`), that window contains only 7 limit-hit days (1.29%), 1 full-day lock (0.185%), and 10 tail events exceeding 3σ, so $(n, \bar{\lambda}) = (0.3, 0.04)$ and $(p_{\mathrm{up}}, \eta_{\mathrm{up}}, \eta_{\mathrm{dn}}) = (0.4, 40.0, 25.0)$ are fixed as pre-registered scenario values rather than joint maximum-likelihood estimates; because all evaluations are conducted in a semi-synthetic simulator partially calibrated to GFEX `LC` futures rather than on empirical option market prices, the external validity of our conclusions has not been tested on real market data. Second, our causal threshold filter $\hat{I}_t = \mathbb{I}(|r_t| > 3\sigma \vee \mathrm{limit_hit}_t = 1)$ misses moderate jumps with $|J| < 3\sigma = 0.0484$, resulting in a true-positive recall of 13.78% on `TH2b` (TP $= 4,138$, FP $= 1,757$, precision 70.20%, raw detection ratio 19.63%, bias -45.17%) and 18.63% on `S4_test.npz` (TP $= 44,052$, FP $= 4,831$, precision 90.12%, $r = 0.4455$, bias -24.44%). Third, when hedging solely with linear futures at baseline cost $c = 10^{-4}$, aggregate CVaR_{95} across all CVaR policies is compressed within 0.021021–0.021187, and `DH_lband` (0.021187) incurs a 0.52% higher CVaR_{95} than unconstrained `DH1` (0.021078) because $1\times$ fee savings (2.56×10^{-5}) are smaller than band tracking error, requiring $6\times$ transaction costs (0.023145 vs. 0.023241, $p = 0.218$) or a 2D option leg to eliminate this penalty. Fourth, in the 10-seed 2D option-hedging benchmark, feeding $\hat{\lambda}_t$ into an unconstrained MLP (`DH_opt1`, 0.015226 ± 0.001497) yields an 11.91% point reduction over `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\mathrm{holm}} = 0.450$, $\mathrm{MDE}_{80\%} = 28.76\%$) that does not achieve standalone statistical significance due to cross-seed optimization variance and collinearity with nonlinear σ_{EWMA} features (`DH_opt_ewma_nl`, 0.015904), and this unconstrained gain attenuates from +11.91% ($\kappa_{\mathrm{iv}} = 0$) to +5.31% ($\kappa_{\mathrm{iv}} = 2$) and +0.73% ($\kappa_{\mathrm{iv}} = 5$) under dynamic implied-volatility spikes, so our 2D percentage gains do not constitute estimates of realizable trading profits. Fifth, off-policy `D4PG-QR` (`C1-C9`) exhibits a 20%–30% seed collapse rate ($\mathrm{CVaR}_{95} > 0.10$)

whenever jump or limit signals are included, with non-collapsed `D4PG-QR (C1)` seeds averaging 0.038934 ± 0.020739 and degrading further at $20,000$ updates. Sixth, while each trajectory's limit ratio ℓ is drawn from the empirical GFEX distribution (7%–15%), ℓ is held constant within each 60-day path rather than modeling dynamic consecutive-day exchange limit adjustments.

5.3 Conclusion

In our semi-synthetic numerical setting partially calibrated to Guangzhou Futures Exchange Lithium Carbonate (`LC`) futures, we conducted a multi-model empirical and microstructural audit of Deep Hedging under self-exciting Hawkes jump clustering and institutional price-limit locks. Our findings support four principal conclusions:

1. **Asymmetric Tail Risk Optimization:** Across all tested market regimes and risk-aversion parameters, training with the Rockafellar-Uryasev CVaR_{95} objective outperforms both classical analytical baselines (`B76-Delta`, `Leland-Delta`, `WW-Band`) and symmetric mean-variance neural policies (`DH_mv`) in the $(\text{CVaR}_{95}, \text{Cost})$ plane, avoiding the empirical tail-risk lower bound and excessive turnover induced by symmetric variance penalties.

2. **Microstructural Adaptation via Jump-Modulated No-Trade Bands and Limit Features:** Embedding the causal online Hawkes intensity $\hat{\lambda}_t$ into a multiplicative no-trade band (`DH_lband`, `DH_full`, and `DH_opt_lband`) leads the network to consistently learn a negative modulation slope ($\hat{\theta} < 0$), narrowing the no-trade buffer during jump-clustering bursts and widening it during calm periods. In the single-instrument setting, this mechanism reduces trading frequency and transaction costs while achieving the lowest CVaR_{99} and conditional locked-subset CVaR_{95}, matches unconstrained `DH1` under elevated transaction costs, and—when combined with price-limit features in `DH_full`—compresses the cumulative unexecuted lock-step hedging gap (`LockGap`) across both the S4 benchmark and zero-shot generalization to the S5 extreme stress regime.

3. **Option-Gamma Spanning and Structural Band Parameterization as the Dual Keys for Jump Filtering:** Input Jacobian saliency diagnostics, nonlinear EWMA controls, and oracle-intensity experiments demonstrate that linear futures alone cannot hedge *ex-ante* quadratic jump-Gamma risk. Extending the action space to include a rolling short-dated ATM call option (`TorchHedgeEnvOpt`) breaks the single-instrument spanning ceiling across all policies, and channeling $\hat{\lambda}_t$ through the multiplicative no-trade band (`DH_opt_lband`) overcomes the cross-seed optimization variance and EWMA collinearity of unconstrained MLPs to deliver a statistically significant and economically large tail-risk reduction.

4. **Algorithmic Advantage of Direct BPTT over Off-Policy Distributional RL:** When the accounting and microstructural execution rules are known and piecewise differentiable, end-to-end BPTT (`DH`) provides zero seed collapse and sub-1% cross-seed variation, outperforming off-policy quantile actor-critic (`D4PG-QR`) across all training budgets.

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